

هوش مصنوعی و کارگاه

دانشکده ریاضی و علوم کامپیوتر دانشگاه صنعتی امیرکبیر



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Artificial Intelligence & Workshop

روشهای خوشه بندی (خوشه بندی مبتنی بر
همسایگی، خوشه بندی سلسله مراتبی، خوشه
بندی مبتنی بر چگالی و ...)

Ref:

Data Mining: Concepts and Techniques

(3rd ed.)

— Chapter 10 —

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Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts 
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods
- Grid-Based Methods
- Evaluation of Clustering
- Summary

What is Cluster Analysis?

- Cluster: A collection of data objects
 - similar (or related) to one another within the same group
 - dissimilar (or unrelated) to the objects in other groups
- Cluster analysis (or *clustering*, *data segmentation*, ...)
 - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- Unsupervised learning: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- Typical applications
 - As a stand-alone tool to get insight into data distribution
 - As a preprocessing step for other algorithms

Clustering as a Preprocessing Tool (Utility)

- **Summarization:**
 - Preprocessing for regression, PCA, classification, and association analysis
- **Compression:**
 - Image processing: vector quantization
- **Finding K-nearest Neighbors**
 - Localizing search to one or a small number of clusters
- **Outlier detection**
 - Outliers are often viewed as those “far away” from any cluster

Quality: What Is Good Clustering?

- A good clustering method will produce high quality clusters
 - high intra-class similarity: **cohesive** within clusters
 - low inter-class similarity: **distinctive** between clusters
- The quality of a clustering method depends on
 - the similarity measure used by the method
 - its implementation, and
 - Its ability to discover some or all of the hidden patterns

Measure the Quality of Clustering

- Dissimilarity/Similarity metric
 - Similarity is expressed in terms of a distance function, typically metric: $d(i, j)$
 - The definitions of distance functions are usually rather different for interval-scaled, boolean, categorical, ordinal ratio, and vector variables
- Quality of clustering:
 - There is usually a separate “quality” function that measures the “goodness” of a cluster.
 - It is hard to define “similar enough” or “good enough”
 - The answer is typically highly subjective

Considerations for Cluster Analysis

- Partitioning criteria
 - Single level vs. hierarchical partitioning (often, multi-level hierarchical partitioning is desirable)
- Separation of clusters
 - Exclusive (e.g., one customer belongs to only one region) vs. non-exclusive (e.g., one document may belong to more than one class)
- Similarity measure
 - Distance-based (e.g., Euclidian, road network, vector) vs. connectivity-based (e.g., density or contiguity)
- Clustering space
 - Full space (often when low dimensional) vs. subspaces (often in high-dimensional clustering)

Major Clustering Approaches (I)

- Partitioning approach:
 - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: k-means, k-medoids, CLARANS
- Hierarchical approach:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
 - Based on connectivity and density functions
 - Typical methods: DBSACN, OPTICS, DenClue
- Grid-based approach:
 - based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

Major Clustering Approaches (II)

- Model-based:
 - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
 - Typical methods: EM, SOM, COBWEB
- Frequent pattern-based:
 - Based on the analysis of frequent patterns
 - Typical methods: p-Cluster
- User-guided or constraint-based:
 - Clustering by considering user-specified or application-specific constraints
 - Typical methods: COD (obstacles), constrained clustering
- Link-based clustering:
 - Objects are often linked together in various ways
 - Massive links can be used to cluster objects: SimRank, LinkClus

Similarity and Dissimilarity

- **Similarity**
 - Numerical measure of how alike two data objects are
 - Value is higher when objects are more alike
 - Often falls in the range $[0,1]$
- **Dissimilarity** (e.g., distance)
 - Numerical measure of how different two data objects are
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies
- **Proximity** refers to a similarity or dissimilarity

Data Matrix and Dissimilarity Matrix

- Data matrix

- n data points with p dimensions
- Two modes

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix

- n data points, but registers only the distance
- A triangular matrix

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

Proximity Measure for Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- Method 1: Simple matching
 - m : # of matches, p : total # of variables
$$d(i, j) = \frac{p - m}{p}$$
- Method 2: Use a large number of binary attributes
 - creating a new binary attribute for each of the M nominal states

Proximity Measure for Binary Attributes

- A contingency table for binary data

		Object j		
		1	0	sum
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p

- Distance measure for symmetric binary variables:
- Distance measure for asymmetric binary variables:
- Jaccard coefficient (*similarity* measure for *asymmetric* binary variables):

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

$$d(i, j) = \frac{r + s}{q + r + s}$$

$$sim_{Jaccard}(i, j) = \frac{q}{q + r + s}$$

Distance on Numeric Data: Minkowski Distance

- *Minkowski distance*: A popular distance measure

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and h is the order (the distance so defined is also called L- h norm)

- **Properties**
 - $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positive definiteness)
 - $d(i, j) = d(j, i)$ (Symmetry)
 - $d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality)
- A distance that satisfies these properties is a **metric**

Special Cases of Minkowski Distance

- $h = 1$: Manhattan (city block, L_1 norm) distance
 - E.g., the Hamming distance: the number of bits that are different between two binary vectors

$$d(i, j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \dots + |x_{i_p} - x_{j_p}|$$

- $h = 2$: (L_2 norm) Euclidean distance

$$d(i, j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + \dots + |x_{i_p} - x_{j_p}|^2)}$$

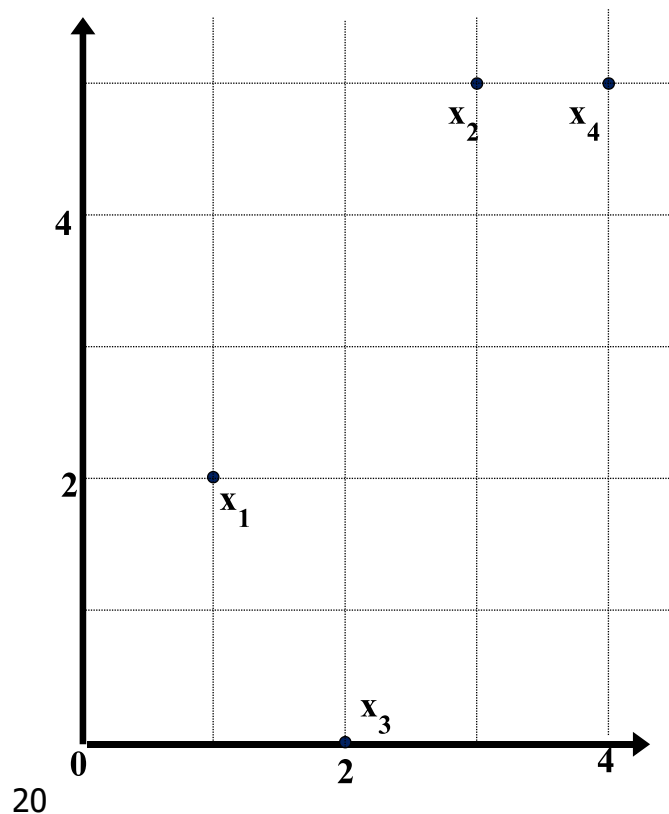
- $h \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.
 - This is the maximum difference between any component (attribute) of the vectors

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|$$

Example: Minkowski Distance

Dissimilarity Matrices

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Supremum

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.

<i>Document</i>	<i>team</i>	<i>coach</i>	<i>hockey</i>	<i>baseball</i>	<i>soccer</i>	<i>penalty</i>	<i>score</i>	<i>win</i>	<i>loss</i>	<i>season</i>
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Other vector objects: gene features in micro-arrays, ...
- Applications: information retrieval, biologic taxonomy, gene feature mapping, ...
- Cosine measure: If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| ||d_2||,$$

where $\bullet$ indicates vector dot product, $||d||$: the length of vector d

Example: Cosine Similarity

- $\cos(d_1, d_2) = (d_1 \bullet d_2) / (||d_1|| ||d_2||)$,
where $\bullet$ indicates vector dot product, $||d||$: the length of vector d
- Ex: Find the **similarity** between documents 1 and 2.

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \bullet d_2 = 5*3 + 0*0 + 3*2 + 0*0 + 2*1 + 0*1 + 0*1 + 2*1 + 0*0 + 0*1 = 25$$

$$||d_1|| = (5*5 + 0*0 + 3*3 + 0*0 + 2*2 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} = 6.481$$

$$||d_2|| = (3*3 + 0*0 + 2*2 + 0*0 + 1*1 + 1*1 + 0*0 + 1*1 + 0*0 + 1*1)^{0.5} = (17)^{0.5} = 4.12$$

$$\cos(d_1, d_2) = 0.94$$

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Partitioning Algorithms: Basic Concept

- Partitioning method: Partitioning a database D of n objects into a set of k clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

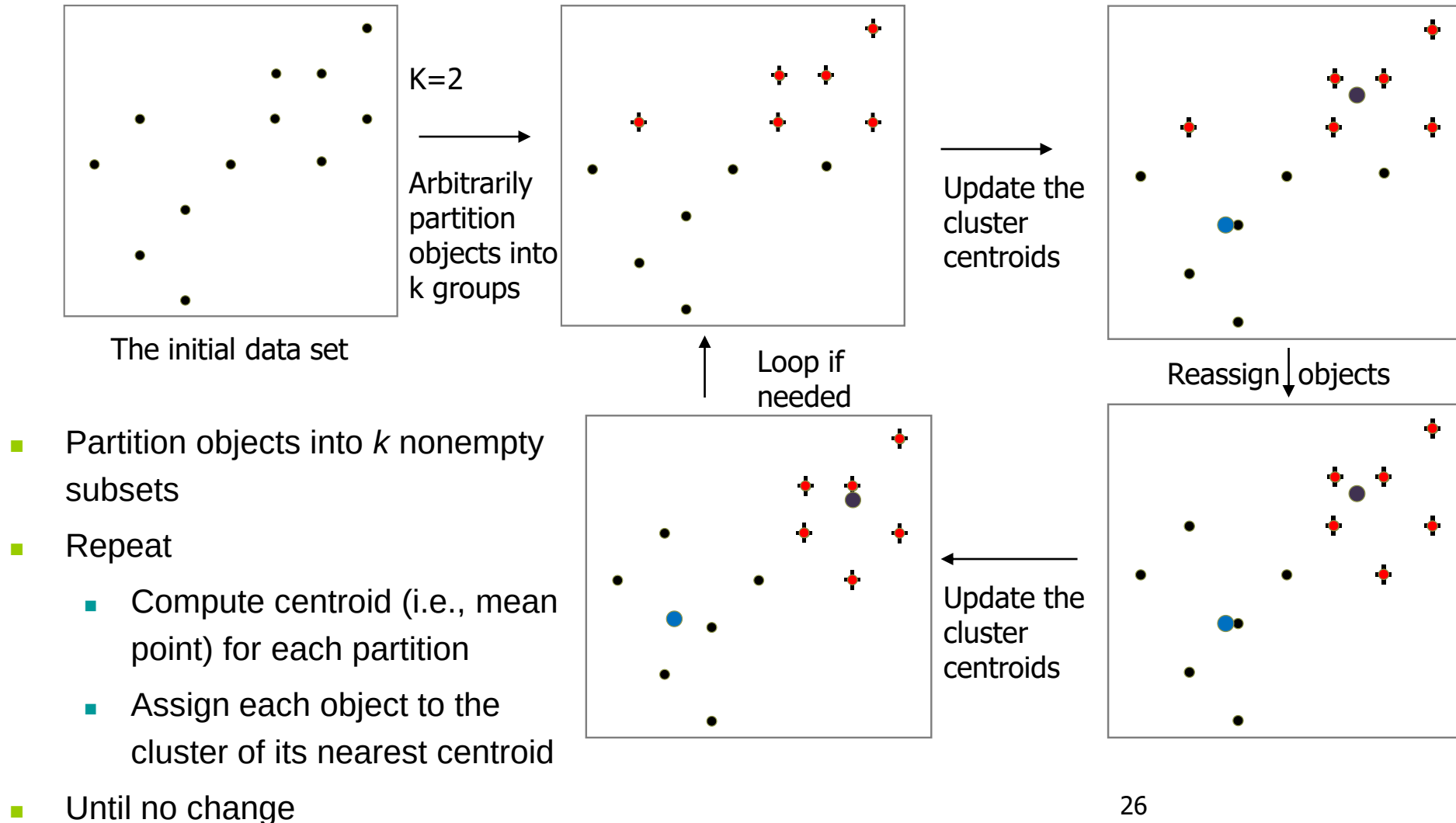
$$E = \sum_{i=1}^k \sum_{p \in C_i} (p - c_i)^2$$

- Given k , find a partition of k clusters that optimizes the chosen partitioning criterion

The *K-Means* Clustering Method

- Given k , the *k-means* algorithm is implemented in four steps:
 - Partition objects into k nonempty subsets
 - Compute seed points as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., *mean point*, of the cluster)
 - Assign each object to the cluster with the nearest seed point
 - Go back to Step 2, stop when the assignment does not change

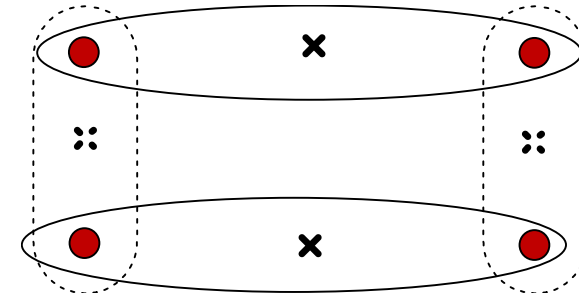
An Example of *K-Means* Clustering



Variations of the *K-Means* Method

- Most of the variants of the *k-means* which differ in

- Selection of the initial k means
- Dissimilarity calculations
- Strategies to calculate cluster means

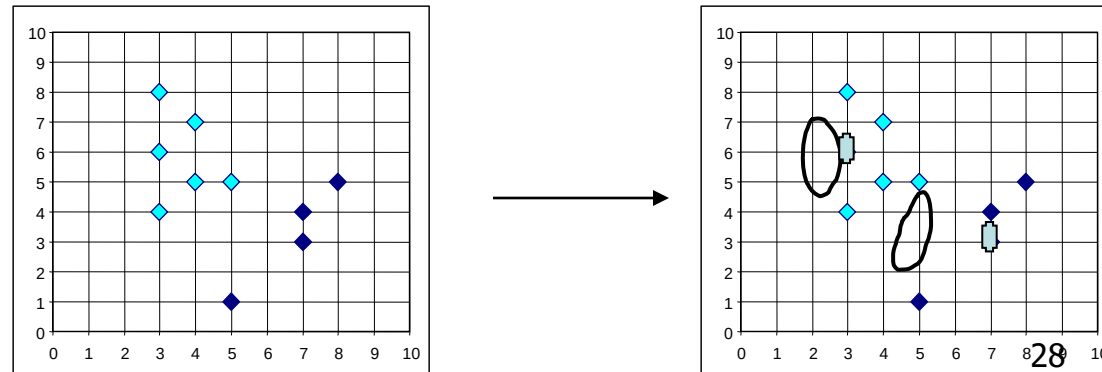


- Handling categorical data: *k-modes*

- Replacing means of clusters with modes
- Using new dissimilarity measures to deal with categorical objects
- Using a frequency-based method to update modes of clusters
- A mixture of categorical and numerical data: *k-prototype* method

What Is the Problem of the K-Means Method?

- The k-means algorithm is sensitive to outliers !
 - Since an object with an extremely large value may substantially distort the distribution of the data
- K-Medoids: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster



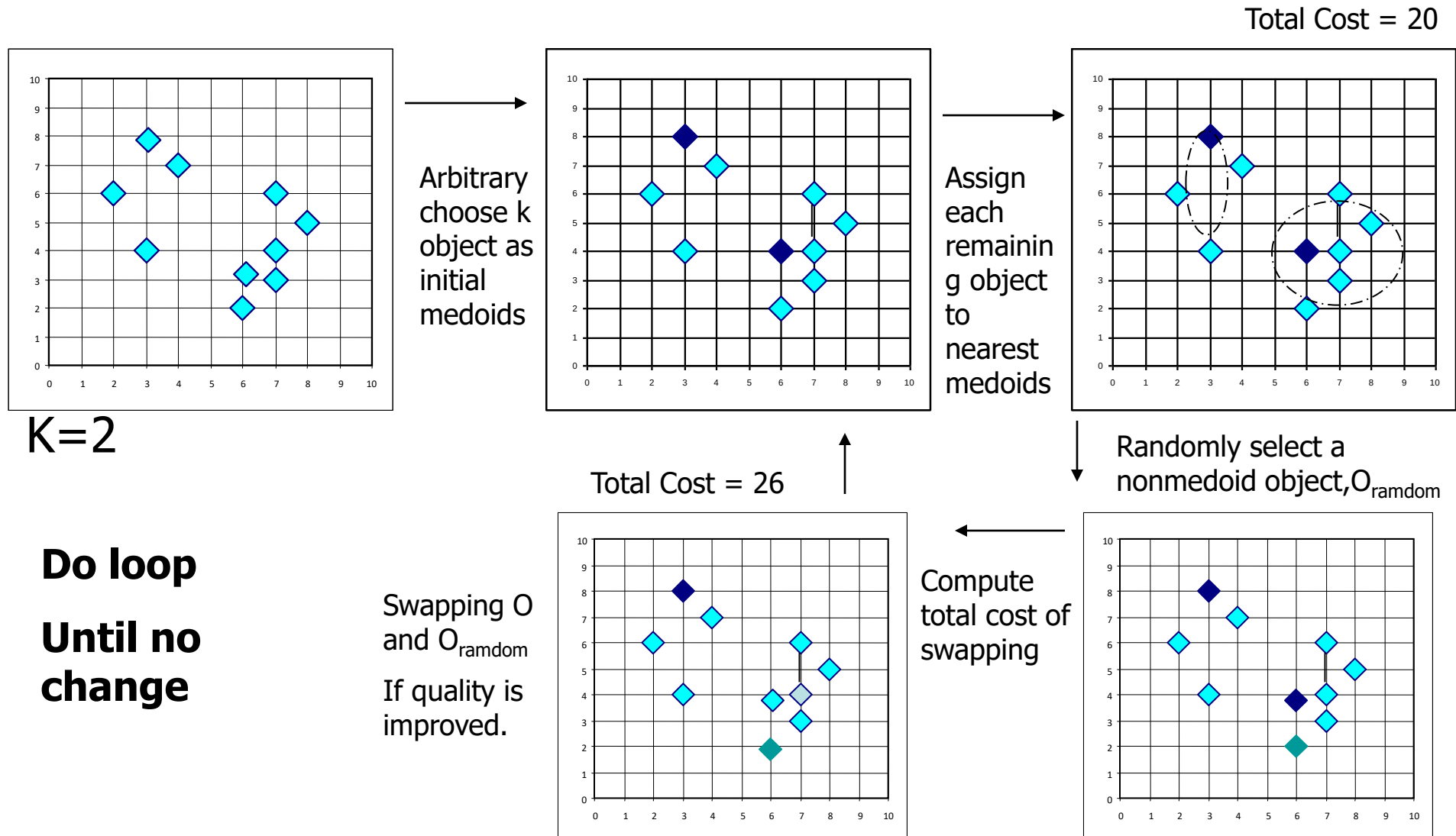
The K-Medoid Clustering Method

- *K-Medoids Clustering*: Find *representative* objects (medoids) in clusters
 - *PAM* (Partitioning Around Medoids, Kaufmann & Rousseeuw 1987)
 - Starts from an initial set of medoids and iteratively replaces one of the medoids by one of the non-medoids if it improves the total distance of the resulting clustering
 - *PAM* works effectively for small data sets, but does not scale well for large data sets (due to the computational complexity)
- *Efficiency improvement on PAM*
 - *CLARA* (Kaufmann & Rousseeuw, 1990): *PAM* on samples
 - *CLARANS* (Ng & Han, 1994): Randomized re-sampling

PAM (Partitioning Around Medoids)

1. INPUT:
2. $D = n \times n$ dissimilarity matrix
3. k = number of clusters
4. // PHASE 1: BUILD
5. $M = \text{select_k_initial_medoids}(D, k)$
6. // PHASE 2: SWAP
7. converged = False
8. while not converged:
9. assign each point to nearest medoid in M
10. TD = total cost of this configuration
11. best_swap_cost = 0
12. best_swap = (None, None)
13. for each medoid m in M :
14. for each non-medoid x :
15. $S = 0$ // Cost change for swap (m, x)
16. for each data point j :
17. // Calculate the change in cost for point j if we swap m with x
18. $S += \text{calculate_cost_change}(j, m, x)$
19. if $S < \text{best_swap_cost}$:
20. best_swap_cost = S
21. best_swap = (m, x)
22. if best_swap_cost < 0:
23. perform the swap: $M = (M \setminus \{\text{best_swap.m}\}) \cup \{\text{best_swap.x}\}$
24. else:
25. converged = True
26. OUTPUT:
27. Set of final medoids M and cluster assignments.

PAM: A Typical K-Medoids Algorithm



CLARA (Clustering LARge Applications) - Sampling Approach

- **Core Idea**
- Instead of applying PAM to the entire dataset, CLARA:
 1. Draws multiple random samples from the full dataset
 2. Applies full PAM to each sample to find medoids
 3. Returns the best clustering found across all samples

CLARANS (Clustering Large Applications based on RANdomized Search) - Graph Search Approach

1. Initialize: Select random node (random k medoids)
2. Random Walk:
 1. Randomly select a neighbor node (swap one medoid)
3. Calculate cost change for entire dataset
4. If cost decreases, move to that neighbor
5. Repeat for maxneighbor iterations without improvement
6. Restart: Start new random walk from different initial node
7. Terminate: After numlocal restarts, return best solution found

■ Parameters

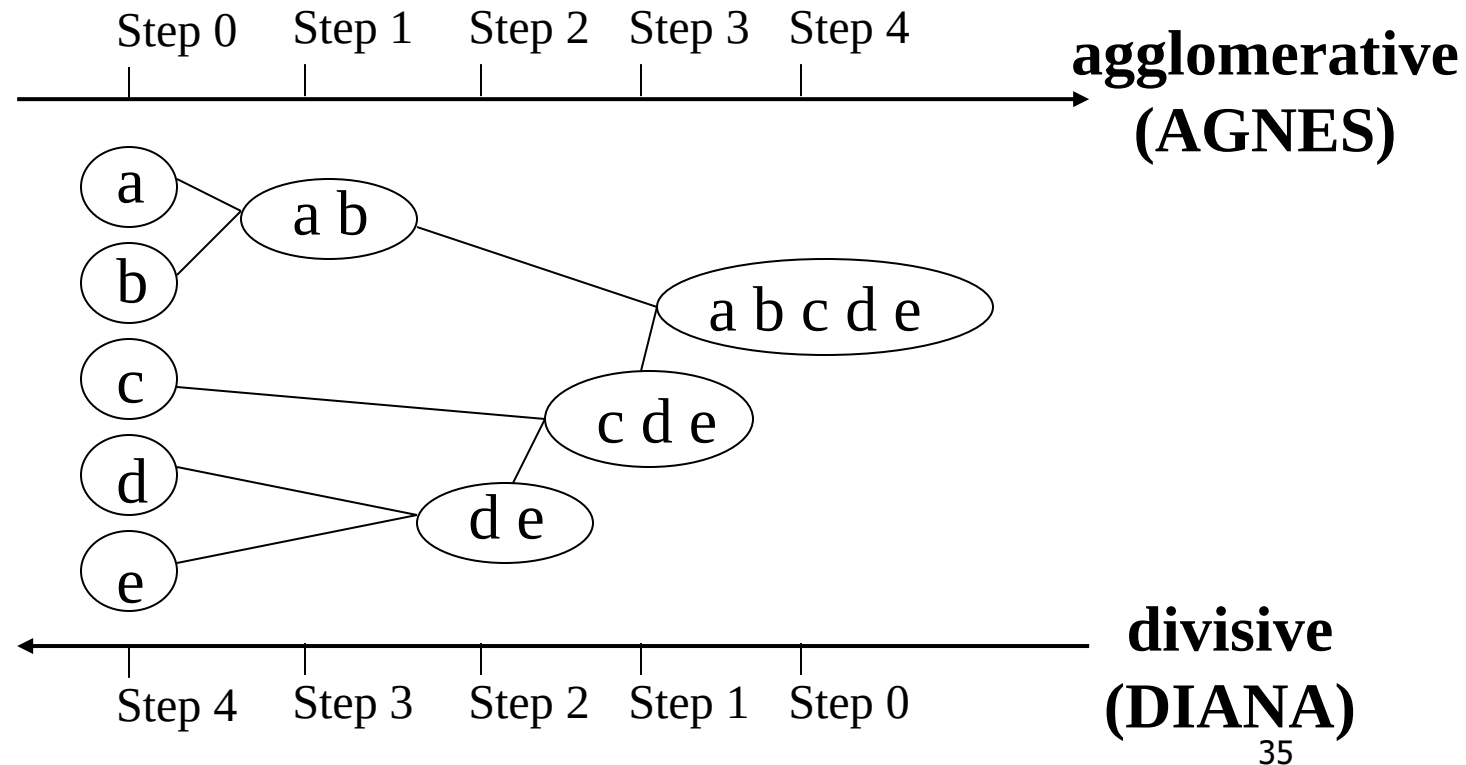
- maxneighbor: Number of neighbors to examine before restarting (typically 250)
- numlocal: Number of restarts (typically 5)

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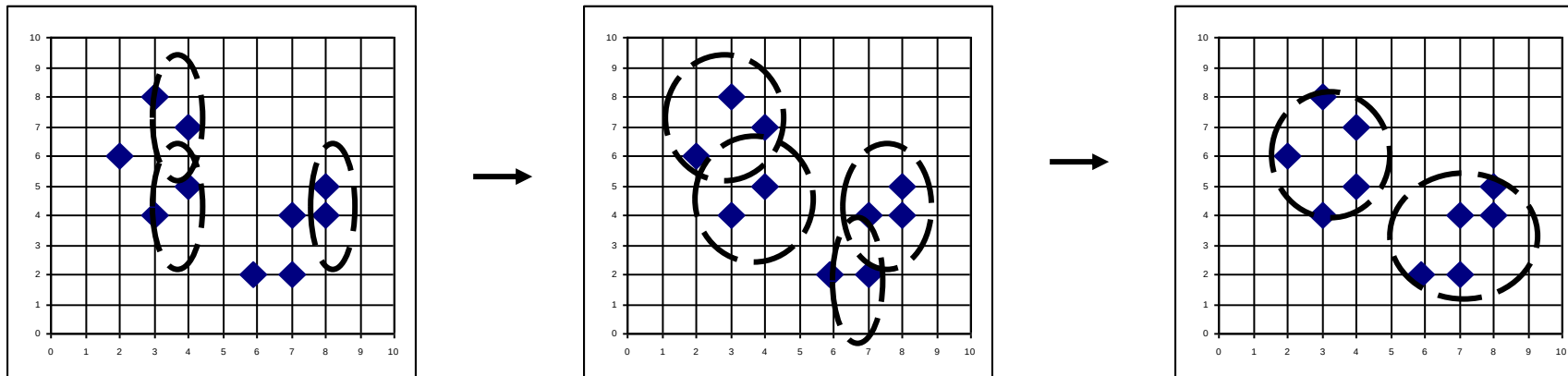
Hierarchical Clustering

- Use distance matrix as clustering criteria. This method does not require the number of clusters k as an input, but needs a termination condition

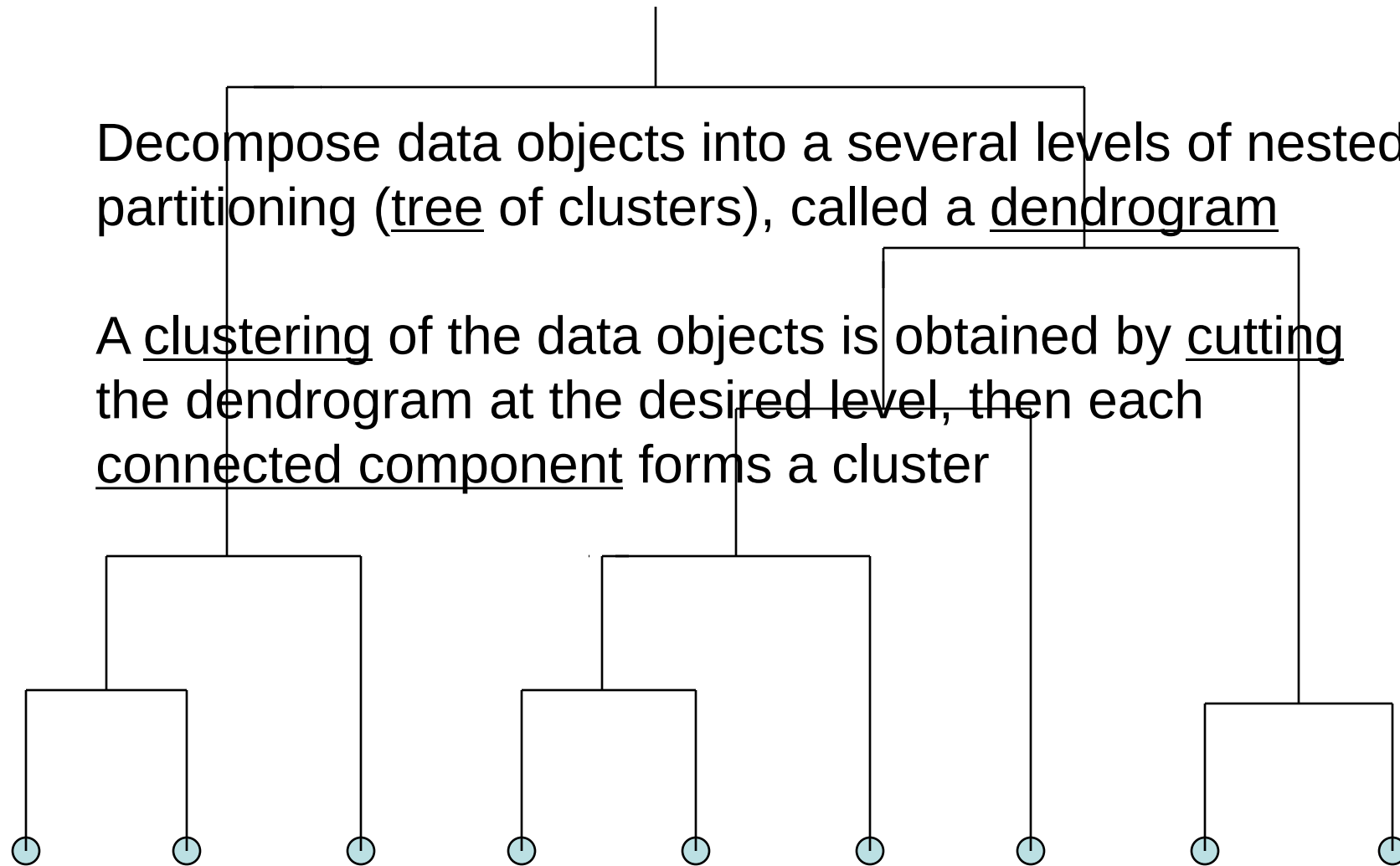


AGNES (Agglomerative Nesting)

- Introduced in Kaufmann and Rousseeuw (1990)
- Implemented in statistical packages, e.g., Splus
- Use the **single-link** method and the dissimilarity matrix
- Merge nodes that have the least dissimilarity
- Go on in a non-descending fashion
- Eventually all nodes belong to the same cluster

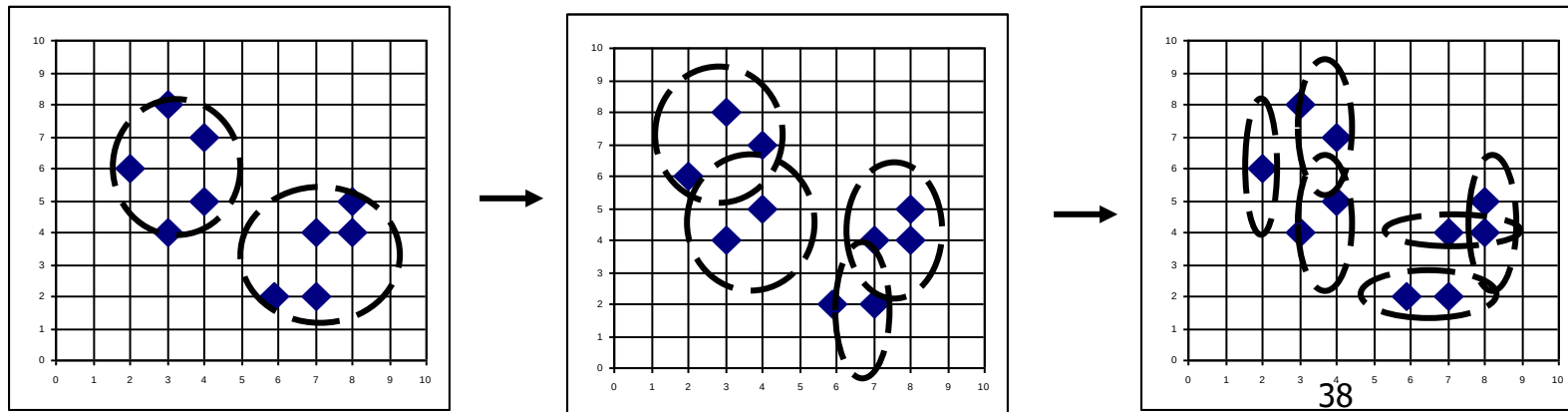


Dendrogram: Shows How Clusters are Merged

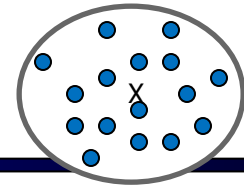
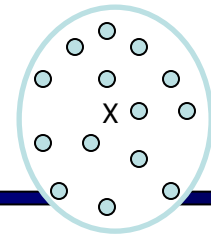


DIANA (Divisive Analysis)

- Introduced in Kaufmann and Rousseeuw (1990)
- Implemented in statistical analysis packages, e.g., Splus
- Inverse order of AGNES
- Eventually each node forms a cluster on its own



Distance between Clusters



- Single link: smallest distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \min(t_{ip}, t_{jq})$
- Complete link: largest distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \max(t_{ip}, t_{jq})$
- Average: avg distance between an element in one cluster and an element in the other, i.e., $\text{dist}(K_i, K_j) = \text{avg}(t_{ip}, t_{jq})$
- Centroid: distance between the centroids of two clusters, i.e., $\text{dist}(K_i, K_j) = \text{dist}(C_i, C_j)$
- Medoid: distance between the medoids of two clusters, i.e., $\text{dist}(K_i, K_j) = \text{dist}(M_i, M_j)$
 - Medoid: a chosen, centrally located object in the cluster

Centroid, Radius and Diameter of a Cluster (for numerical data sets)

- Centroid: the “middle” of a cluster ‘p’
$$C_m = \frac{\sum_{i=1}^N (t_{ip})}{N}$$

- Radius: square root of average distance from any point of the cluster to its centroid

$$R_m = \sqrt{\frac{\sum_{i=1}^N (t_{ip} - c_m)^2}{N}}$$

- Diameter: square root of average mean squared distance between all pairs of points in the cluster

$$D_m = \sqrt{\frac{\sum_{i=1, \dots, N} \sum_{j=1, \dots, N} (t_{ip} - t_{jp})^2}{N(N-1)}}$$

Extensions to Hierarchical Clustering

- Major weakness of agglomerative clustering methods
 - Can never undo what was done previously
 - Do not scale well: time complexity of at least $O(n^2)$, where n is the number of total objects
- Integration of hierarchical & distance-based clustering
 - BIRCH (1996): uses CF-tree and incrementally adjusts the quality of sub-clusters
 - CHAMELEON (1999): hierarchical clustering using dynamic modeling

BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

- Zhang, Ramakrishnan & Livny, SIGMOD'96
- Incrementally construct a CF (Clustering Feature) tree, a hierarchical data structure for multiphase clustering
 - Phase 1: scan DB to build an initial in-memory CF tree (a multi-level compression of the data that tries to preserve the inherent clustering structure of the data)
 - Phase 2: use an arbitrary clustering algorithm to cluster the leaf nodes of the CF-tree
- *Scales linearly*: finds a good clustering with a single scan and improves the quality with a few additional scans
- *Weakness*: handles only numeric data, and sensitive to the order of the data record

Clustering Feature Vector in BIRCH

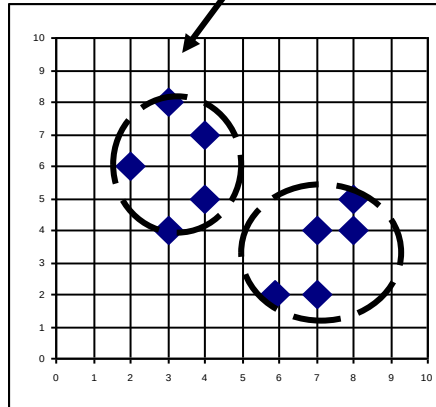
Clustering Feature (CF): $CF = (N, LS, SS)$

N : Number of data points

LS : linear sum of N points: $\sum_{i=1}^N X_i$

SS : square sum of N points

$$\sum_{i=1}^N X_i^2$$



$CF = (5, (16,30), (54,190))$

(3,4)

(2,6)

(4,5)

(4,7)

(3,8)

cluster's centroid, $\mathbf{x}$, radius, R , and diameter, D

$$\mathbf{x}_0 = \frac{\sum_{i=1}^n \mathbf{x}_i}{n} = \frac{LS}{n},$$

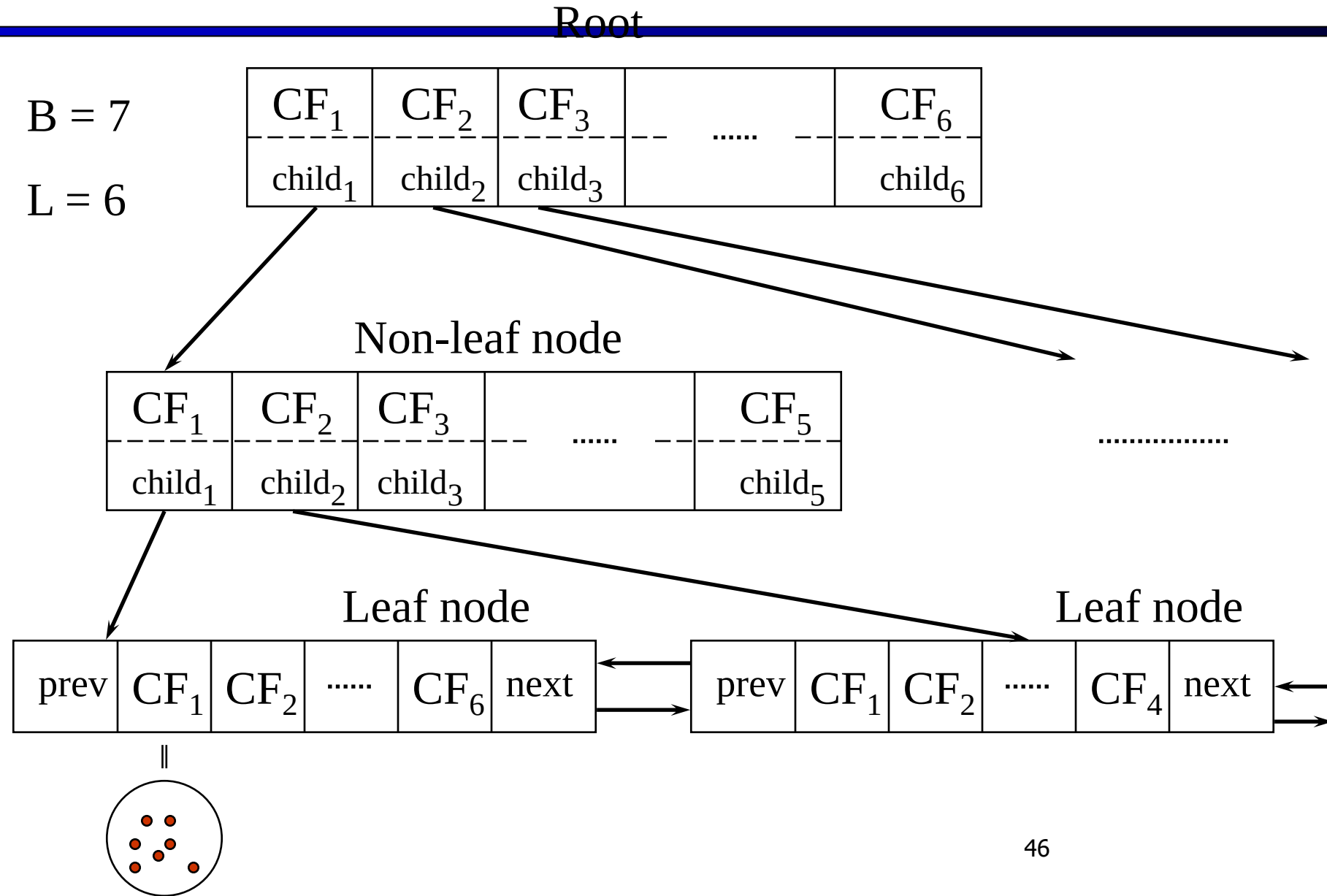
$$R = \sqrt{\frac{\sum_{i=1}^n (\mathbf{x}_i - \mathbf{x}_0)^2}{n}} = \sqrt{\frac{nSS - 2LS^2 + nLS}{n^2}},$$

$$D = \sqrt{\frac{\sum_{i=1}^n \sum_{j=1}^n (\mathbf{x}_i - \mathbf{x}_j)^2}{n(n-1)}} = \sqrt{\frac{2nSS - 2LS^2}{n(n-1)}}.$$

CF-Tree in BIRCH

- Clustering feature:
 - Summary of the statistics for a given subcluster: the 0-th, 1st, and 2nd moments of the subcluster from the statistical point of view
 - Registers crucial measurements for computing cluster and utilizes storage efficiently
- A CF tree is a height-balanced tree that stores the clustering features for a hierarchical clustering
 - A nonleaf node in a tree has descendants or “children”
 - The nonleaf nodes store sums of the CFs of their children
- A CF tree has two parameters
 - Branching factor: max # of children
 - Threshold: max diameter of sub-clusters stored at the leaf nodes

The CF Tree Structure



The Birch Algorithm

- Cluster Diameter

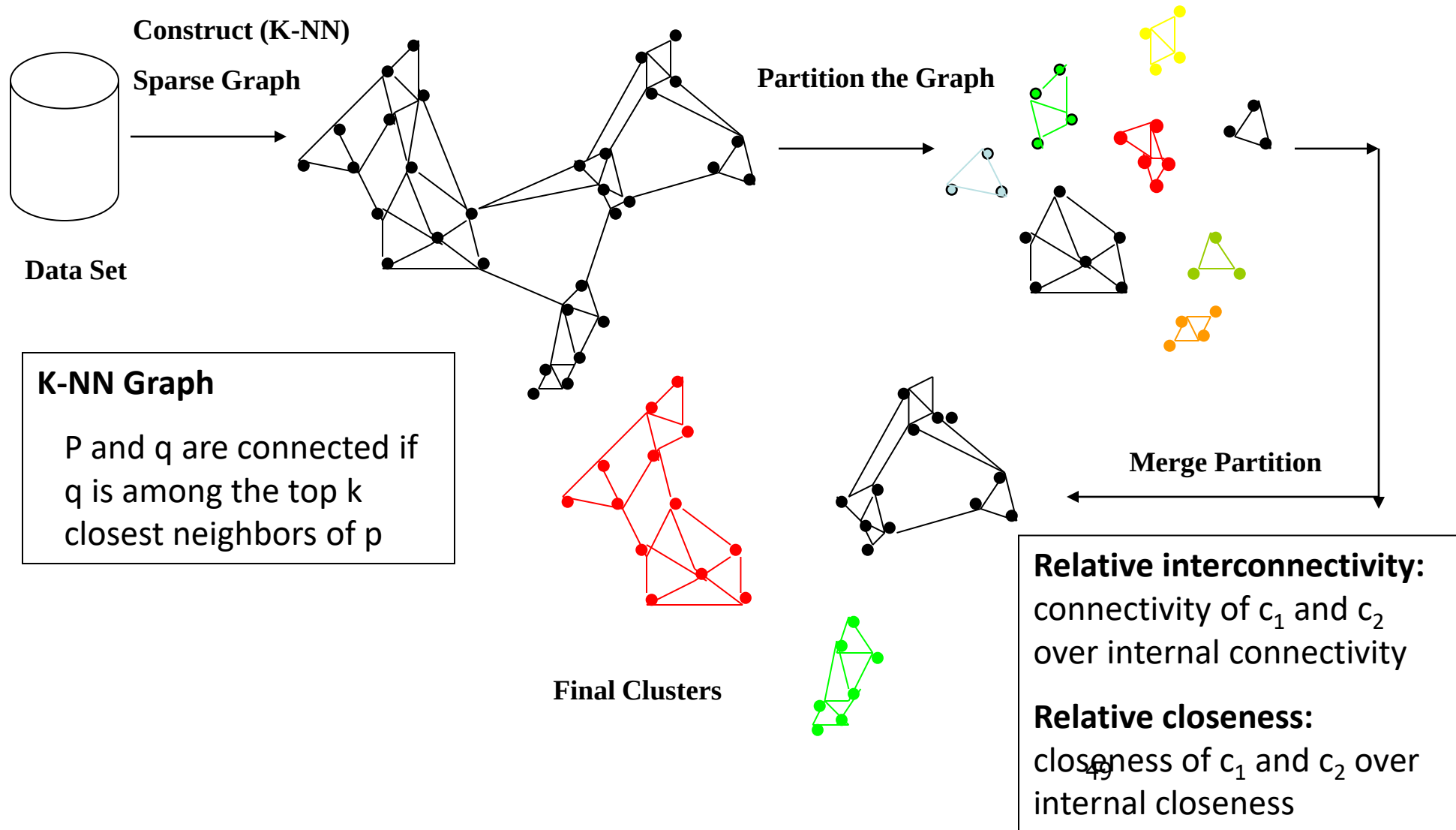
$$\sqrt{\frac{1}{n(n-1)} \sum (x_i - x_j)^2}$$

- For each point in the input
 - Find closest leaf entry
 - Add point to leaf entry and update CF
 - If entry diameter > max_diameter, then split leaf, and possibly parents
- Algorithm is $O(n)$
- Concerns
 - Sensitive to insertion order of data points
 - Since we fix the size of leaf nodes, so clusters may not be so natural
 - Clusters tend to be spherical given the radius and diameter measures

CHAMELEON: Hierarchical Clustering Using Dynamic Modeling (1999)

- CHAMELEON: G. Karypis, E. H. Han, and V. Kumar, 1999
- Measures the similarity based on a dynamic model
 - Two clusters are merged only if the *interconnectivity* and *closeness (proximity)* between two clusters are high *relative to* the internal interconnectivity of the clusters and closeness of items within the clusters
- Graph-based, and a two-phase algorithm
 1. Use a graph-partitioning algorithm: cluster objects into a large number of relatively small sub-clusters
 2. Use an agglomerative hierarchical clustering algorithm: find the genuine clusters by repeatedly combining these sub-clusters

Overall Framework of CHAMELEON



Relative interconnectivity and Relative closeness

- The **relative interconnectivity**, $RI(C_i, C_j)$, between two clusters, C_i and C_j , is defined as the absolute interconnectivity between C_i and C_j , normalized with respect to the internal interconnectivity of the two clusters, C_i and C_j . That is,

$$RI(C_i, C_j) = \frac{|EC_{\{C_i, C_j\}}|}{\frac{1}{2}(|EC_{C_i}| + |EC_{C_j}|)}, \quad (10.12)$$

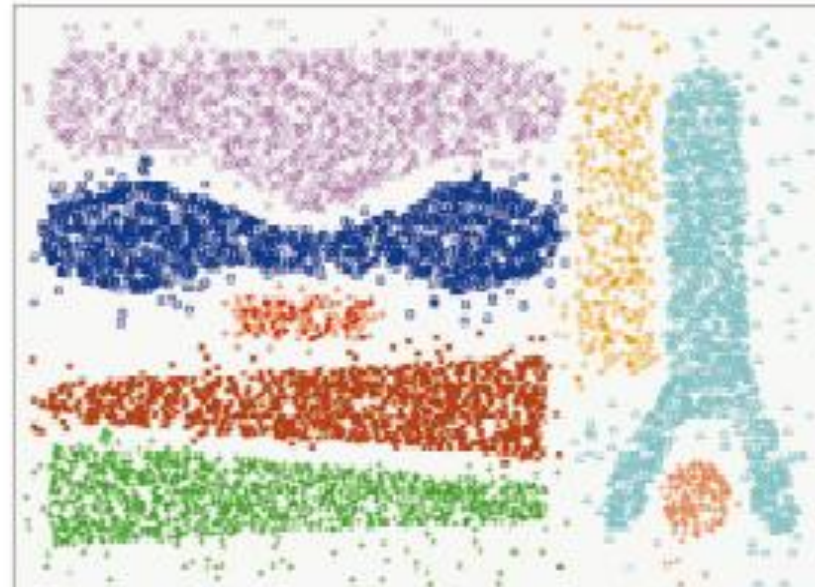
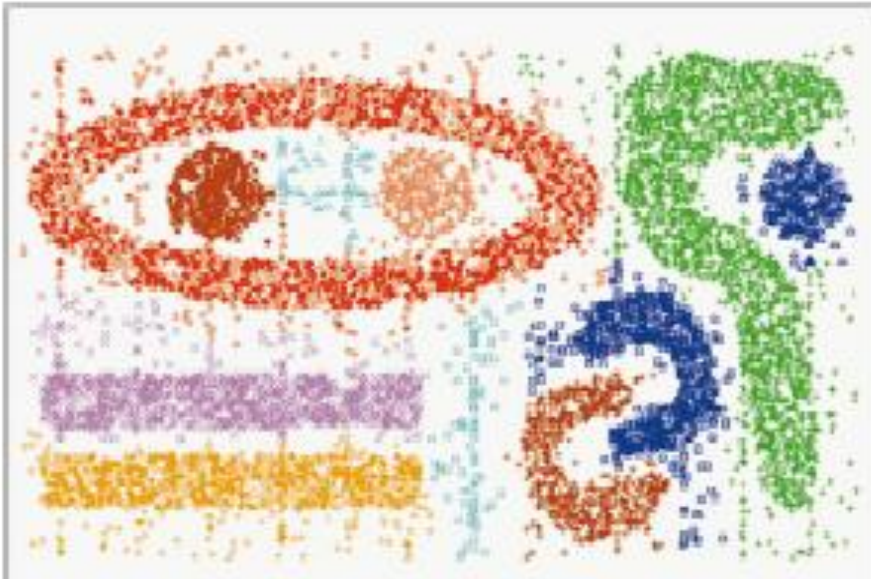
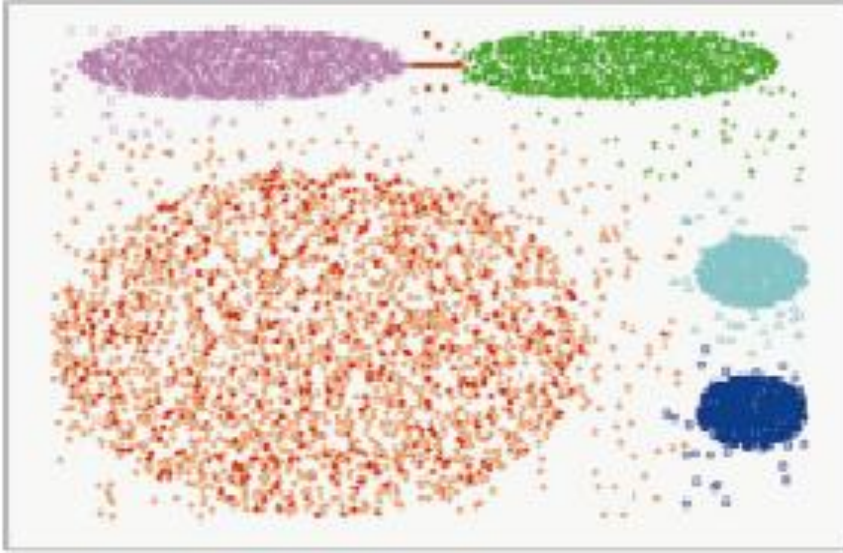
where $EC_{\{C_i, C_j\}}$ is the edge cut as previously defined for a cluster containing both C_i and C_j . Similarly, EC_{C_i} (or EC_{C_j}) is the minimum sum of the cut edges that partition C_i (or C_j) into two roughly equal parts.

- The **relative closeness**, $RC(C_i, C_j)$, between a pair of clusters, C_i and C_j , is the absolute closeness between C_i and C_j , normalized with respect to the internal closeness of the two clusters, C_i and C_j . It is defined as

$$RC(C_i, C_j) = \frac{\bar{S}_{EC_{\{C_i, C_j\}}}}{\frac{|C_i|}{|C_i|+|C_j|} \bar{S}_{EC_{C_i}} + \frac{|C_j|}{|C_i|+|C_j|} \bar{S}_{EC_{C_j}}}, \quad (10.13)$$

where $\bar{S}_{EC_{\{C_i, C_j\}}}$ is the average weight of the edges that connect vertices in C_i to vertices in C_j , and $\bar{S}_{EC_{C_i}}$ (or $\bar{S}_{EC_{C_j}}$) is the average weight of the edges that belong to the min-cut bisector of cluster C_i (or C_j).

CHAMELEON (Clustering Complex Objects)



Chapter 10. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods 
- Grid-Based Methods
- Evaluation of Clustering
- Summary

Density-Based Clustering Methods

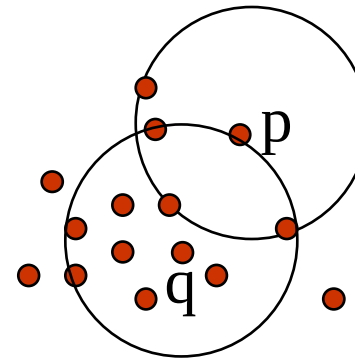
- Clustering based on density (local cluster criterion), such as density-connected points
- Major features:
 - Discover clusters of arbitrary shape
 - Handle noise
 - One scan
 - Need density parameters as termination condition
- Several interesting studies:
 - DBSCAN: Ester, et al. (KDD'96)
 - OPTICS: Ankerst, et al (SIGMOD'99).
 - DENCLUE: Hinneburg & D. Keim (KDD'98)
 - CLIQUE: Agrawal, et al. (SIGMOD'98) (more grid-based)

Density-Based Clustering: Basic Concepts

- Two parameters:
 - *Eps*: Maximum radius of the neighbourhood
 - *MinPts*: Minimum number of points in an *Eps*-neighbourhood of that point
- $N_{Eps}(p)$: $\{q \text{ belongs to } D \mid \text{dist}(p,q) \leq Eps\}$
- **Directly density-reachable**: A point p is directly density-reachable from a point q w.r.t. Eps , $MinPts$ if

- p belongs to $N_{Eps}(q)$
- core point condition:

$$|N_{Eps}(q)| \geq MinPts$$



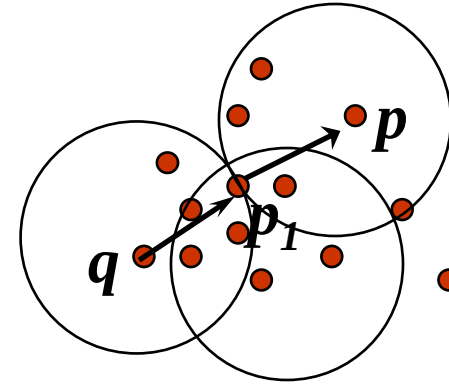
MinPts = 5

Eps = 1 cm

Density-Reachable and Density-Connected

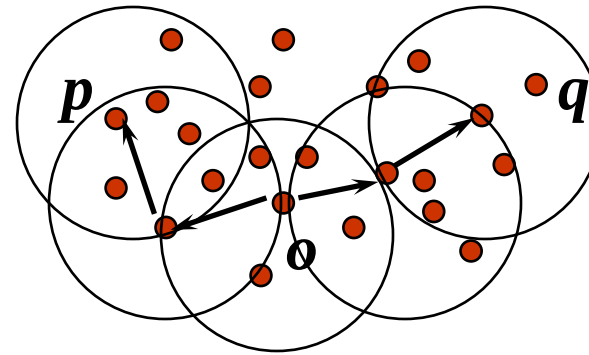
- Density-reachable:

- A point p is **density-reachable** from a point q w.r.t. $Eps, MinPts$ if there is a chain of points $p_1, \dots, p_n$, $p_1 = q, p_n = p$ such that p_{i+1} is directly density-reachable from p_i



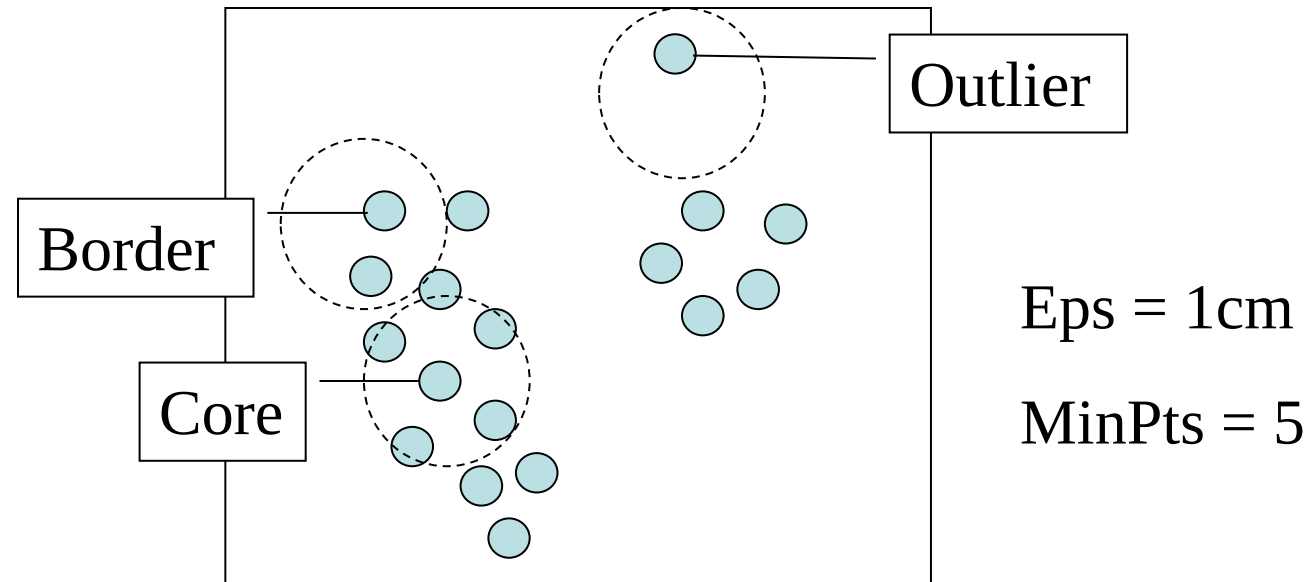
- Density-connected

- A point p is **density-connected** to a point q w.r.t. $Eps, MinPts$ if there is a point o such that both, p and q are density-reachable from o w.r.t. Eps and $MinPts$



DBSCAN: Density-Based Spatial Clustering of Applications with Noise

- Relies on a *density-based* notion of cluster: A *cluster* is defined as a maximal set of density-connected points
- Discovers clusters of arbitrary shape in spatial databases with noise



DBSCAN: The Algorithm

- Arbitrary select a point p
- Retrieve all points density-reachable from p w.r.t. Eps and $MinPts$
- If p is a core point, a cluster is formed
- If p is a border point, no points are density-reachable from p and DBSCAN visits the next point of the database
- Continue the process until all of the points have been processed

DBSCAN: Drawbacks

- It encumbers users with the responsibility of selecting parameter values that will lead to the discovery of acceptable clusters.
- Such parameter settings are usually empirically set and difficult to determine, especially for real-world, high dimensional data sets.
- Most algorithms are sensitive to these parameter values: Slightly different settings may lead to very different clusterings of the data.
- Moreover, real-world, high-dimensional data sets often have very skewed distributions such that their intrinsic clustering structure may not be well characterized by a single set of *global* density parameters.

DBSCAN: Sensitive to Parameters

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

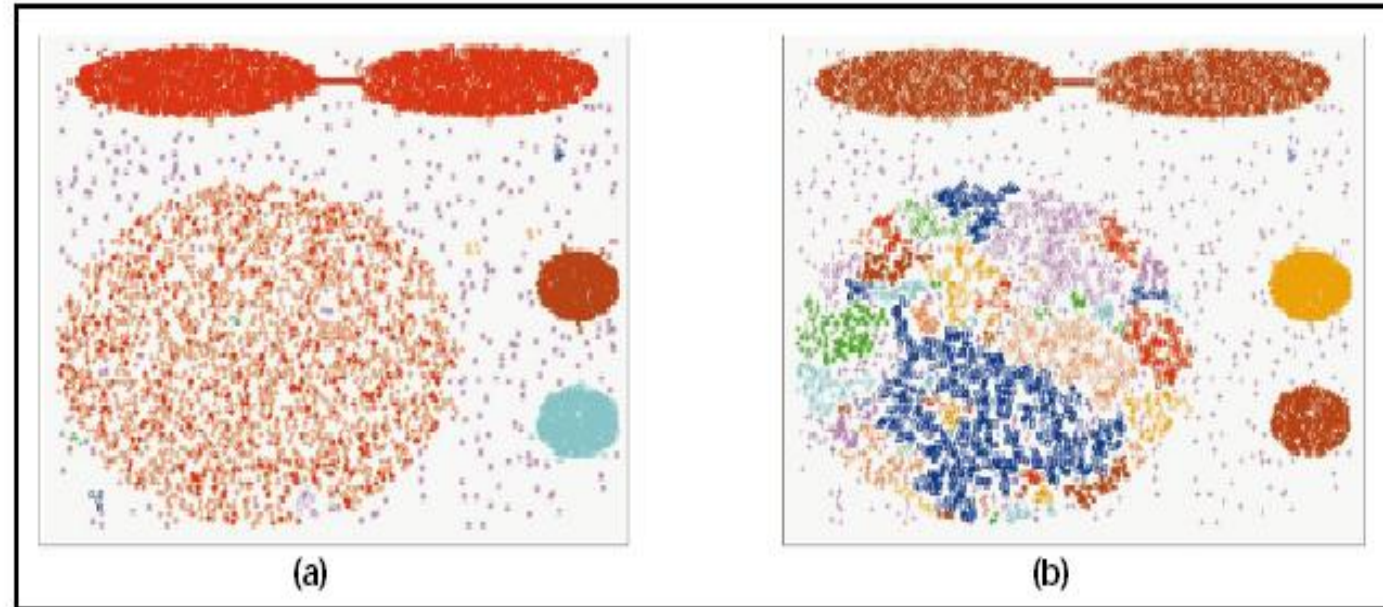
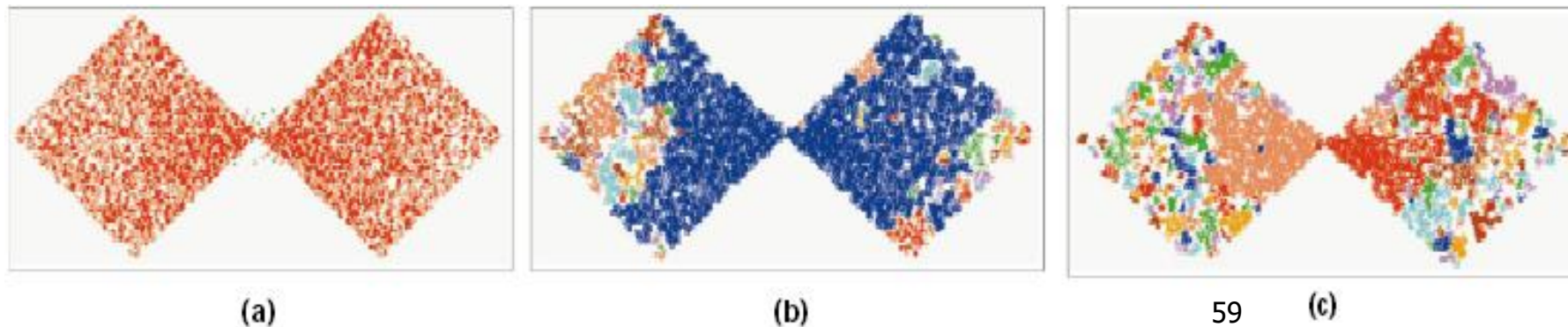


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.



OPTICS: A Cluster-Ordering Method (1999)

- OPTICS: Ordering Points To Identify the Clustering Structure
 - Ankerst, Breunig, Kriegel, and Sander (SIGMOD'99)
 - Produces a special order of the database wrt its density-based clustering structure
 - This cluster-ordering contains info equiv to the density-based clusterings corresponding to a broad range of parameter settings
 - Good for both automatic and interactive cluster analysis, including finding intrinsic clustering structure
 - Can be represented graphically or using visualization techniques

OPTICS

- To overcome the difficulty in using one set of global parameters in clustering analysis, a cluster analysis method called **OPTICS** was proposed.
- OPTICS does not explicitly produce a data set clustering. Instead, it outputs a **cluster ordering**.
- This is a linear list of all objects under analysis and represents the *density-based clustering structure* of the data.
- Objects in a denser cluster are listed closer to each other in the cluster ordering.
- This ordering is equivalent to density-based clustering obtained from a wide range of parameter settings.
- Thus, OPTICS does not require the user to provide a specific density threshold.
- The cluster ordering can be used to extract basic clustering information (e.g., cluster centers, or arbitrary-shaped clusters), derive the intrinsic clustering structure, as well as provide a visualization of the clustering.

OPTICS: Main Approach

- To construct the different clusterings simultaneously, the objects are processed in a specific order.
- This order selects an object that is density-reachable with respect to the lowest ε value so that clusters with higher density (lower ε) will be finished first.
- Based on this idea, OPTICS needs two important pieces of information per object:

OPTICS: Main Approach ...

- The **core-distance** of an object p is the smallest value ε such that the ε -neighborhood of p has at least $MinPts$ objects. That is, ε is the minimum distance threshold that makes p a core object.
- The **reachability-distance** to object p from q is the minimum radius value that makes p density-reachable from q . According to the definition of density-reachability, q has to be a core object and p must be in the neighborhood of q . Therefore, the reachability-distance from q to p is $\max\{core-distance(q), dist(p, q)\}$.

OPTICS: Main Approach ...

- An object p may be directly reachable from multiple core objects. Therefore, p may have multiple reachability-distances with respect to different core objects. The smallest reachability-distance of p is of particular interest because it gives the shortest path for which p is connected to a dense cluster.

OPTICS: Some Extension from DBSCAN

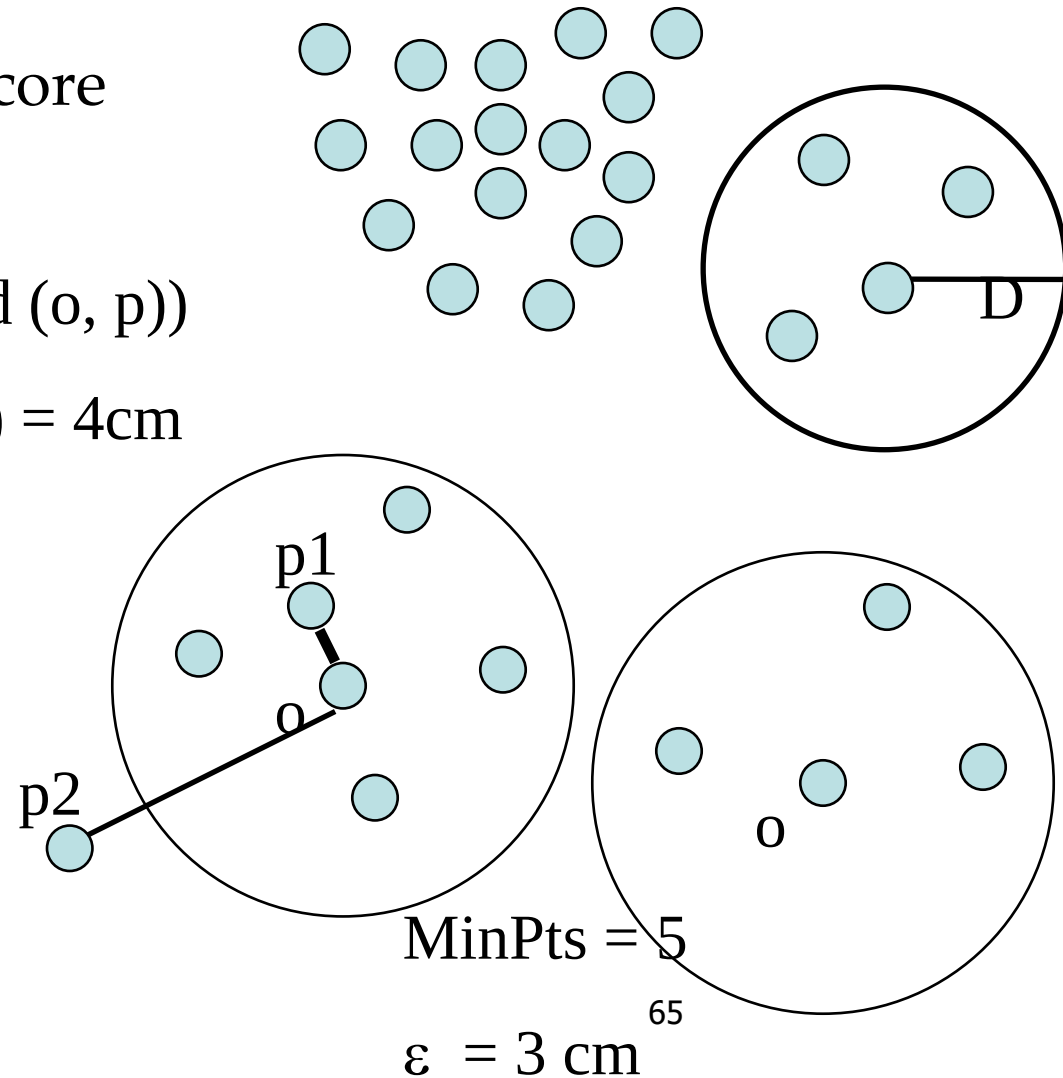
- Core Distance:

- min eps s.t. point is core

- Reachability Distance

$\text{Max}(\text{core-distance}(o), d(o, p))$

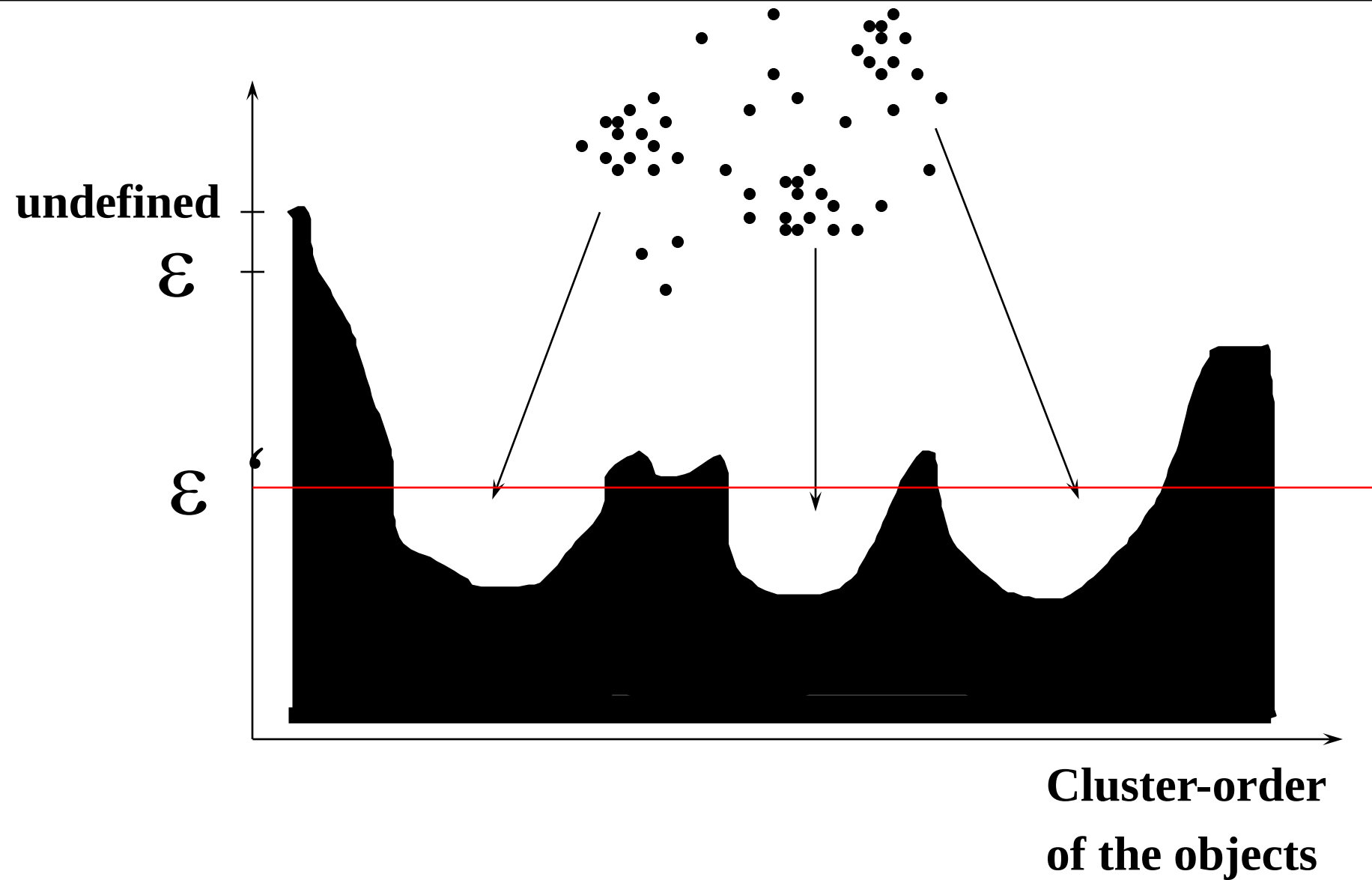
$r(p1, o) = 2.8\text{cm}$. $r(p2, o) = 4\text{cm}$



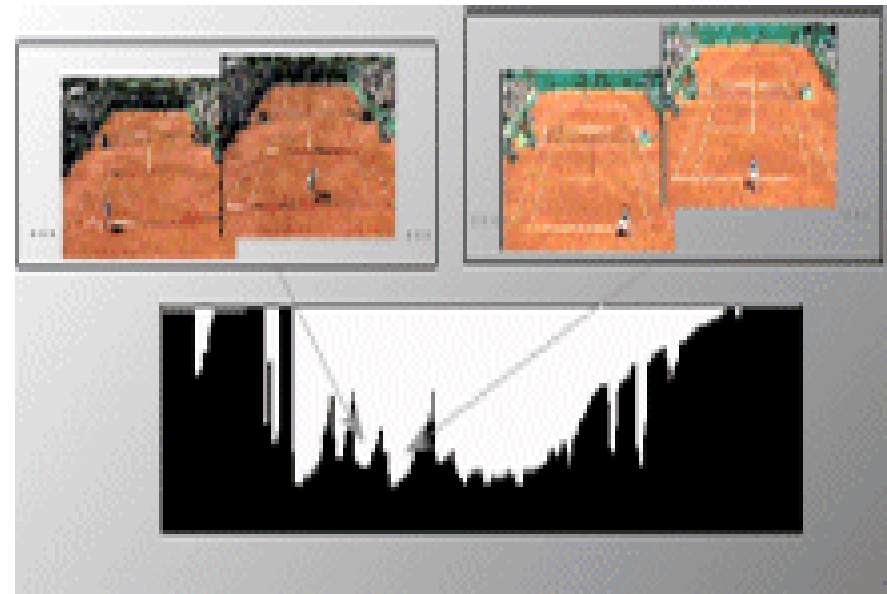
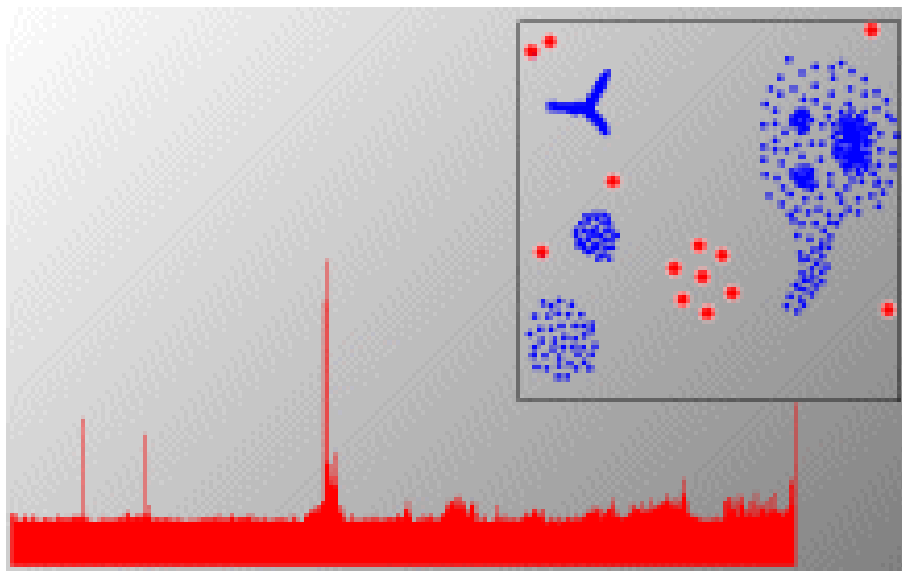
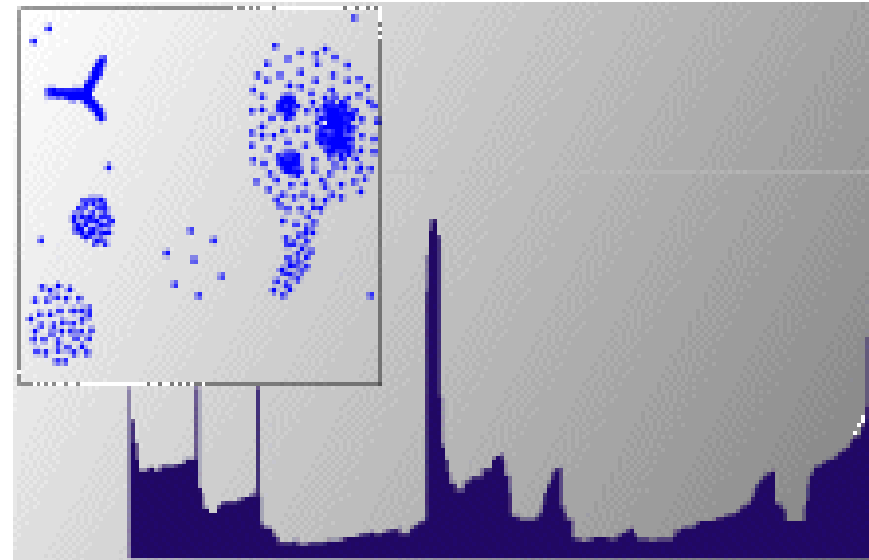
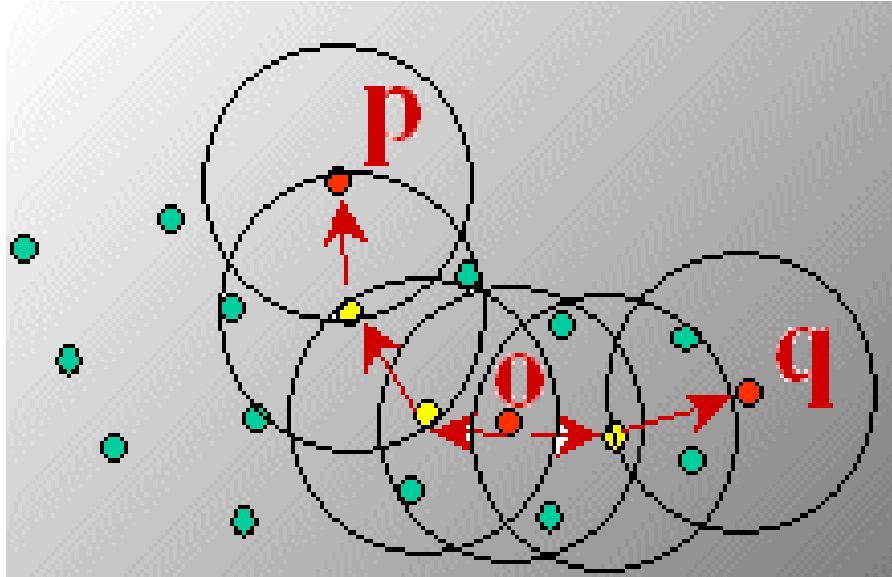
OrderSeeds

- OPTICS computes an ordering of all objects in a given database and, for each object in the database, stores the core-distance and a suitable reachability-distance.
- OPTICS maintains a list called OrderSeeds to generate the output ordering.
- Objects in OrderSeeds are sorted by the reachability-distance from their respective closest core objects, that is, by the smallest reachability-distance of each object.
- OPTICS begins with an arbitrary object from the input database as the current object, p . It retrieves the ε -neighborhood of p , determines the core-distance, and sets the reachability-distance to *undefined*. The current object, p , is then written to output. If p is not a core object, OPTICS simply moves on to the next object in the OrderSeeds list (or the input database if OrderSeeds is empty).

reachability plot for a simple 2-D dataset



Density-Based Clustering: OPTICS & Its Applications



Denclue: Technical Essence

- Density estimation is a core issue in density-based clustering methods. **DENCLUE** (DENsity-based CLUstEring) is a clustering method based on a set of density distribution functions.
- In probability and statistics, **density estimation** is the estimation of an unobservable underlying probability density function based on a set of observed data.
- In the context of density-based clustering, the unobservable underlying probability density function is the true distribution of the population of all possible objects to be analyzed.
- The observed data set is regarded as a random sample from that population.

DENCLUE: Using Statistical Density Functions

- DENsity-based CLUstEring by Hinneburg & Keim (KDD'98)
- Using statistical density functions:

$$f_{\text{Gaussian}}(x, y) = e^{-\frac{d(x, y)^2}{2\sigma^2}}$$

influence of y
on x

$$f_{\text{Gaussian}}^D(x) = \sum_{i=1}^N e^{-\frac{d(x, x_i)^2}{2\sigma^2}}$$

total influence
on x

- Major features

$$\nabla f_{\text{Gaussian}}^D(x, x_i) = \sum_{i=1}^N (x_i - x) \cdot e^{-\frac{d(x, x_i)^2}{2\sigma^2}}$$

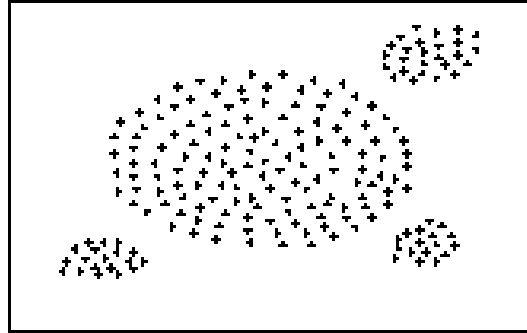
gradient of x in
the direction of
 x_i

- Solid mathematical foundation
- Good for data sets with large amounts of noise
- Allows a compact mathematical description of arbitrarily shaped clusters in high-dimensional data sets
- Significant faster than existing algorithm (e.g., DBSCAN)
- But needs a large number of parameters

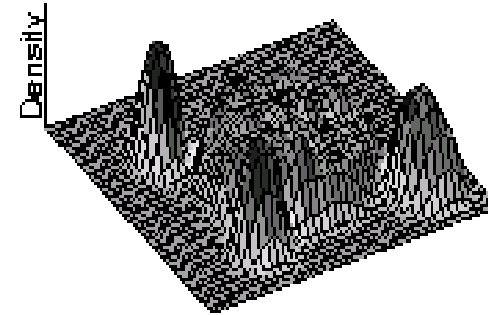
Denclue: Technical Essence

- Uses grid cells but only keeps information about grid cells that do actually contain data points and manages these cells in a tree-based access structure
- Influence function: describes the impact of a data point within its neighborhood
- Overall density of the data space can be calculated as the sum of the influence function of all data points
- Clusters can be determined mathematically by identifying density attractors
- Density attractors are local maximal of the overall density function
- Center defined clusters: assign to each density attractor the points density attracted to it
- Arbitrary shaped cluster: merge density attractors that are connected through paths of high density ($>$ threshold)

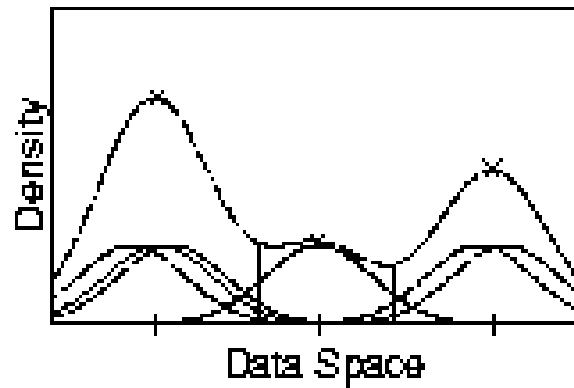
Density Attractor



(a) Data Set



(c) Gaussian



Center-Defined and Arbitrary

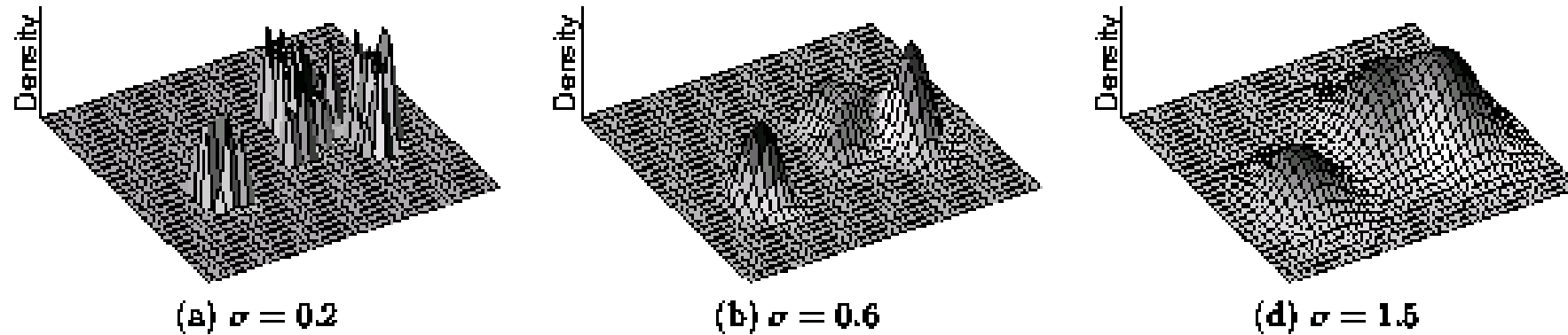


Figure 3: Example of Center-Defined Clusters for different σ

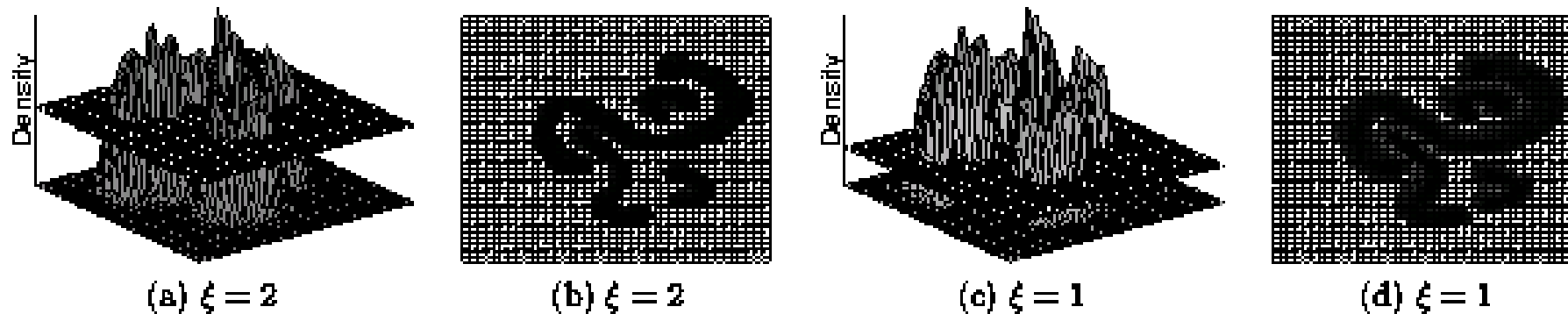


Figure 4: Example of Arbitrary-Shape Clusters for different ξ

Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
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Assessing Clustering Tendency

- Assess if non-random structure exists in the data by measuring the probability that the data is generated by a uniform data distribution
- Test spatial randomness by statistic test: Hopkins Static
 - Given a dataset D regarded as a sample of a random variable o, determine how far away o is from being uniformly distributed in the data space
 - Sample n points, $p_1, \dots, p_n$, uniformly from D. For each p_i , find its nearest neighbor in D: $x_i = \min\{\text{dist}(p_i, v)\}$ where v in D
 - Sample n points, $q_1, \dots, q_n$, uniformly from D. For each q_i , find its nearest neighbor in $D - \{q_i\}$: $y_i = \min\{\text{dist}(q_i, v)\}$ where v in D and $v \neq q_i$
 - Calculate the Hopkins Statistic:
- If D is uniformly distributed, $\sum x_i$ and $\sum y_i$ will be close to each other and H is close to 0.5. If D is highly skewed, H is close to 0

$$H = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i + \sum_{i=1}^n y_i}$$

Determine the Number of Clusters

- Empirical method
 - # of clusters $\approx \sqrt{n/2}$ for a dataset of n points
- Elbow method
 - Use the turning point in the curve of sum of within cluster variance w.r.t the # of clusters
- Cross validation method
 - Divide a given data set into m parts
 - Use $m - 1$ parts to obtain a clustering model
 - Use the remaining part to test the quality of the clustering
 - E.g., For each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and the closest centroids to measure how well the model fits the test set
 - For any $k > 0$, repeat it m times, compare the overall quality measure w.r.t. different k 's, and find # of clusters that fits the data the best

Measuring Clustering Quality

- Two methods: extrinsic vs. intrinsic
- Extrinsic: supervised, i.e., the ground truth is available
 - Compare a clustering against the ground truth using certain clustering quality measure
 - Ex. BCubed precision and recall metrics
- Intrinsic: unsupervised, i.e., the ground truth is unavailable
 - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are
 - Ex. Silhouette coefficient

Measuring Clustering Quality: Extrinsic Methods

- Clustering quality measure: $Q(C, C_g)$, for a clustering C given the ground truth C_g .
- Q is good if it satisfies the following 4 essential criteria
 - Cluster homogeneity: the purer, the better
 - Cluster completeness: should assign objects belong to the same category in the ground truth to the same cluster
 - Rag bag: putting a heterogeneous object into a pure cluster should be penalized more than putting it into a *rag bag* (i.e., “miscellaneous” or “other” category)
 - Small cluster preservation: splitting a small category into pieces is more harmful than splitting a large category into pieces

Summary

- **Cluster analysis** groups objects based on their **similarity** and has wide applications
- Measure of similarity can be computed for **various types of data**
- Clustering algorithms can be **categorized** into partitioning methods, hierarchical methods, density-based methods, grid-based methods, and model-based methods
- **K-means** and **K-medoids** algorithms are popular partitioning-based clustering algorithms
- **Birch** and **Chameleon** are interesting hierarchical clustering algorithms, and there are also probabilistic hierarchical clustering algorithms
- **DBSCAN**, **OPTICS**, and **DENCLU** are interesting density-based algorithms

CS512-Spring 2011: An Introduction

- Coverage
 - Cluster Analysis: Chapter 11
 - Outlier Detection: Chapter 12
 - Mining Sequence Data: BK2: Chapter 8
 - Mining Graphs Data: BK2: Chapter 9
 - Social and Information Network Analysis
 - BK2: Chapter 9
 - Partial coverage: Mark Newman: “Networks: An Introduction”, Oxford U., 2010
 - Scattered coverage: Easley and Kleinberg, “Networks, Crowds, and Markets: Reasoning About a Highly Connected World”, Cambridge U., 2010
 - Recent research papers
 - Mining Data Streams: BK2: Chapter 8
- Requirements
 - One research project
 - One class presentation (15 minutes)
 - Two homeworks (no programming assignment)
 - Two midterm exams (no final exam)

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References (2)

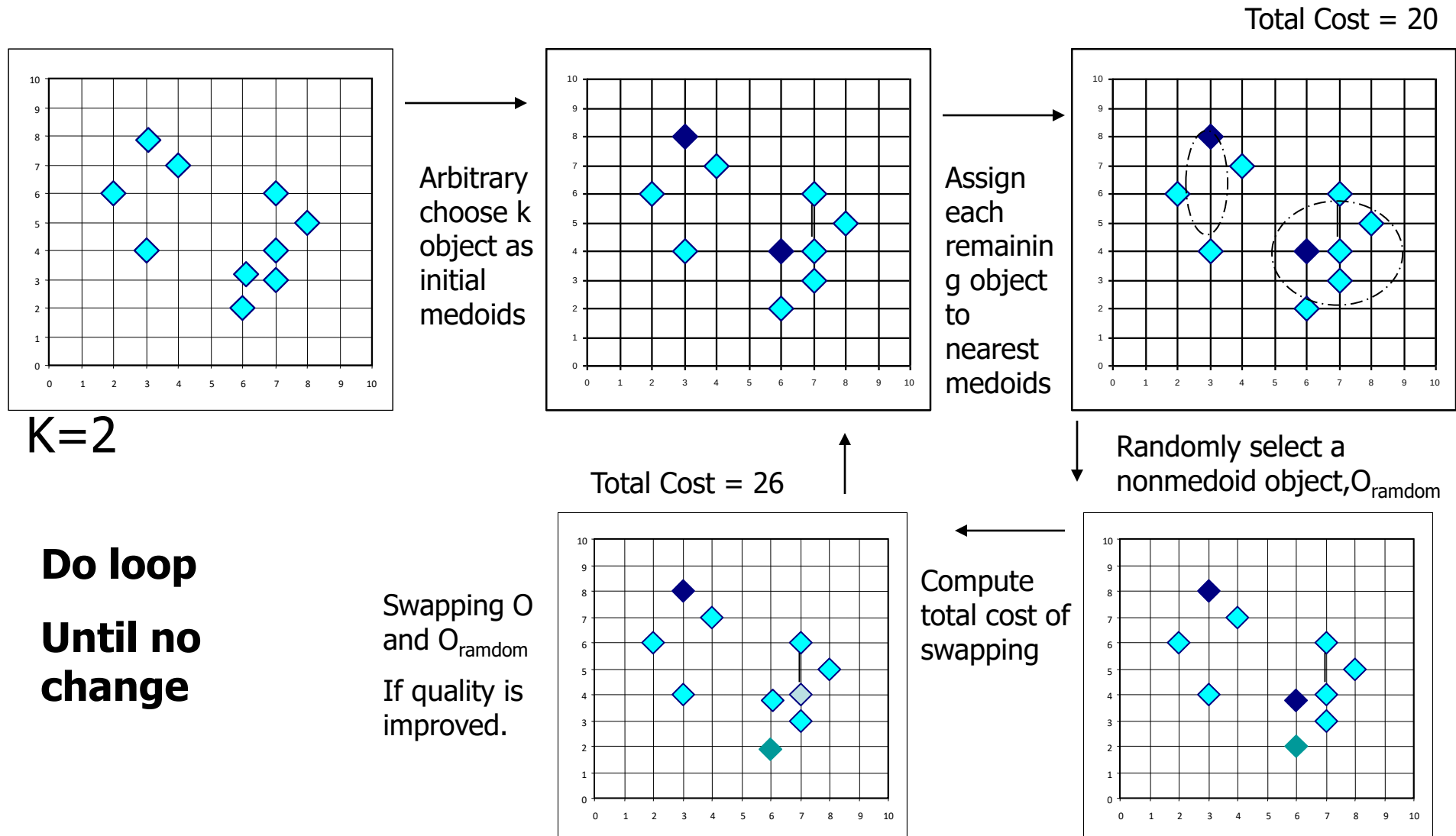
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Slides unused in class

A Typical K-Medoids Algorithm (PAM)

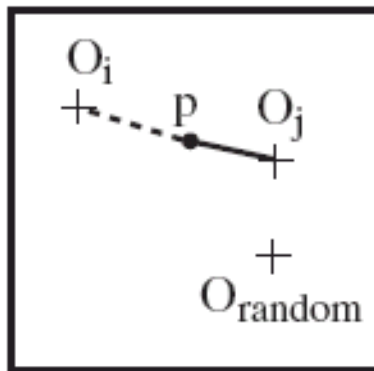


PAM (Partitioning Around Medoids) (1987)

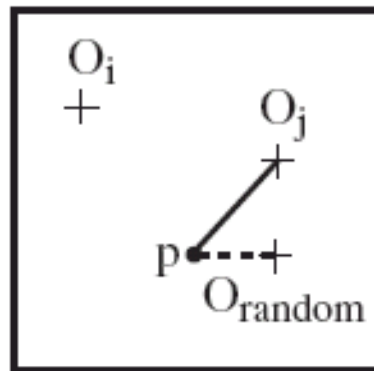
- PAM (Kaufman and Rousseeuw, 1987), built in Splus
- Use real object to represent the cluster
 - Select k representative objects arbitrarily
 - For each pair of non-selected object h and selected object i , calculate the total swapping cost TC_{ih}
 - For each pair of i and h ,
 - If $TC_{ih} < 0$, i is replaced by h
 - Then assign each non-selected object to the most similar representative object
 - repeat steps 2-3 until there is no change

PAM Clustering: Finding the Best Cluster Center

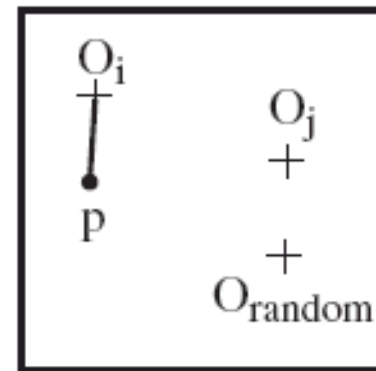
- Case 1: p currently belongs to o_j . If o_j is replaced by o_{random} as a representative object and p is the closest to one of the other representative object o_i , then p is reassigned to o_i



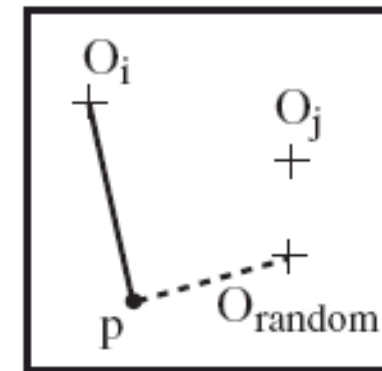
1. Reassigned to O_i



2. Reassigned to O_{random}



3. No change



4. Reassigned to O_{random}

- data object
- + cluster center
- before swapping
- after swapping

What Is the Problem with PAM?

- Pam is more robust than k-means in the presence of noise and outliers because a medoid is less influenced by outliers or other extreme values than a mean
- Pam works efficiently for small data sets but does not **scale well** for large data sets.
 - $O(k(n-k)^2)$ for each iteration

where n is # of data, k is # of clusters

➔ Sampling-based method

CLARA(Clustering LARge Applications)

CLARA (Clustering Large Applications) (1990)

- *CLARA* (Kaufmann and Rousseeuw in 1990)
 - Built in statistical analysis packages, such as SPlus
 - It draws *multiple samples* of the data set, applies *PAM* on each sample, and gives the best clustering as the output
- Strength: deals with larger data sets than *PAM*
- Weakness:
 - Efficiency depends on the sample size
 - A good clustering based on samples will not necessarily represent a good clustering of the whole data set if the sample is biased

CLARANS (“Randomized” CLARA) (1994)

- *CLARANS* (A Clustering Algorithm based on Randomized Search) (Ng and Han'94)
 - Draws sample of neighbors dynamically
 - The clustering process can be presented as searching a graph where every node is a potential solution, that is, a set of k medoids
 - If the local optimum is found, *it* starts with new randomly selected node in search for a new local optimum
- Advantages: More efficient and scalable than both *PAM* and *CLARA*
- Further improvement: Focusing techniques and spatial access structures (Ester et al.'95)

ROCK: Clustering Categorical Data

- ROCK: RObust Clustering using linKs
 - S. Guha, R. Rastogi & K. Shim, ICDE'99
- Major ideas
 - Use links to measure similarity/proximity
 - Not distance-based
- Algorithm: sampling-based clustering
 - Draw random sample
 - Cluster with links
 - Label data in disk
- Experiments
 - Congressional voting, mushroom data

Similarity Measure in ROCK

- Traditional measures for categorical data may not work well, e.g., Jaccard coefficient
- Example: Two groups (clusters) of transactions
 - C_1 . $\langle a, b, c, d, e \rangle$: $\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{c, d, e\}$
 - C_2 . $\langle a, b, f, g \rangle$: $\{a, b, f\}, \{a, b, g\}, \{a, f, g\}, \{b, f, g\}$
- Jaccard co-efficient may lead to wrong clustering result
 - C_1 : 0.2 ($\{a, b, c\}, \{b, d, e\}$) to 0.5 ($\{a, b, c\}, \{a, b, d\}$)
 - C_1 & C_2 : could be as high as 0.5 ($\{a, b, c\}, \{a, b, f\}$)
- Jaccard co-efficient-based similarity function:
$$Sim(T_1, T_2) = \frac{|T_1 \cap T_2|}{|T_1 \cup T_2|}$$
 - Ex. Let $T_1 = \{a, b, c\}, T_2 = \{c, d, e\}$

$$Sim(T_1, T_2) = \frac{|\{c\}|}{|\{a, b, c, d, e\}|} = \frac{1}{5} = 0.2$$

Link Measure in ROCK

- Clusters

- $C_1: \langle a, b, c, d, e \rangle$: $\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{c, d, e\}$
- $C_2: \langle a, b, f, g \rangle$: $\{a, b, f\}, \{a, b, g\}, \{a, f, g\}, \{b, f, g\}$

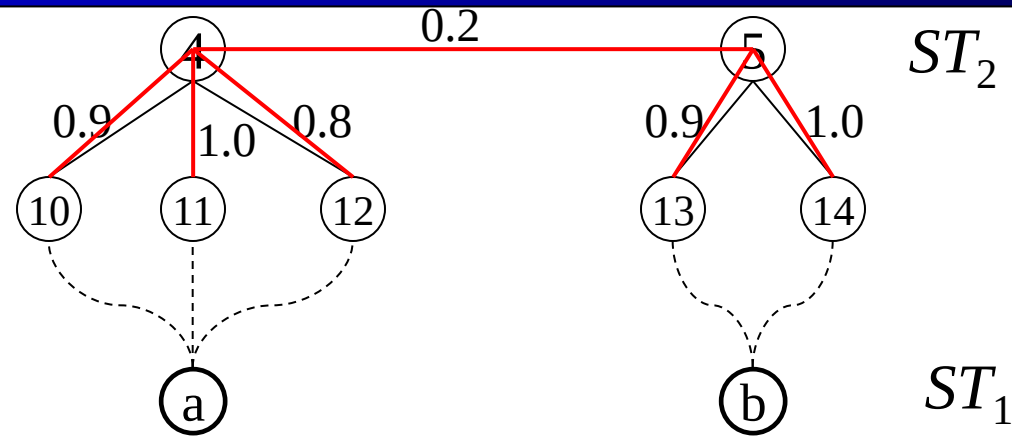
- Neighbors

- Two transactions are neighbors if $\text{sim}(T_1, T_2) > \text{threshold}$
- Let $T_1 = \{a, b, c\}$, $T_2 = \{c, d, e\}$, $T_3 = \{a, b, f\}$
 - T_1 connected to: $\{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{b, c, d\}, \{b, c, e\}, \{a, b, f\}, \{a, b, g\}$
 - T_2 connected to: $\{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, e\}, \{b, d, e\}, \{b, c, d\}$
 - T_3 connected to: $\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, b, g\}, \{a, f, g\}, \{b, f, g\}$

- Link Similarity

- Link similarity between two transactions is the # of common neighbors
- $\text{link}(T_1, T_2) = 4$, since they have 4 common neighbors
 - $\{a, c, d\}, \{a, c, e\}, \{b, c, d\}, \{b, c, e\}$
- $\text{link}(T_1, T_3) = 3$, since they have 3 common neighbors
 - $\{a, b, d\}, \{a, b, e\}, \{a, b, g\}$

Aggregation-Based Similarity Computation



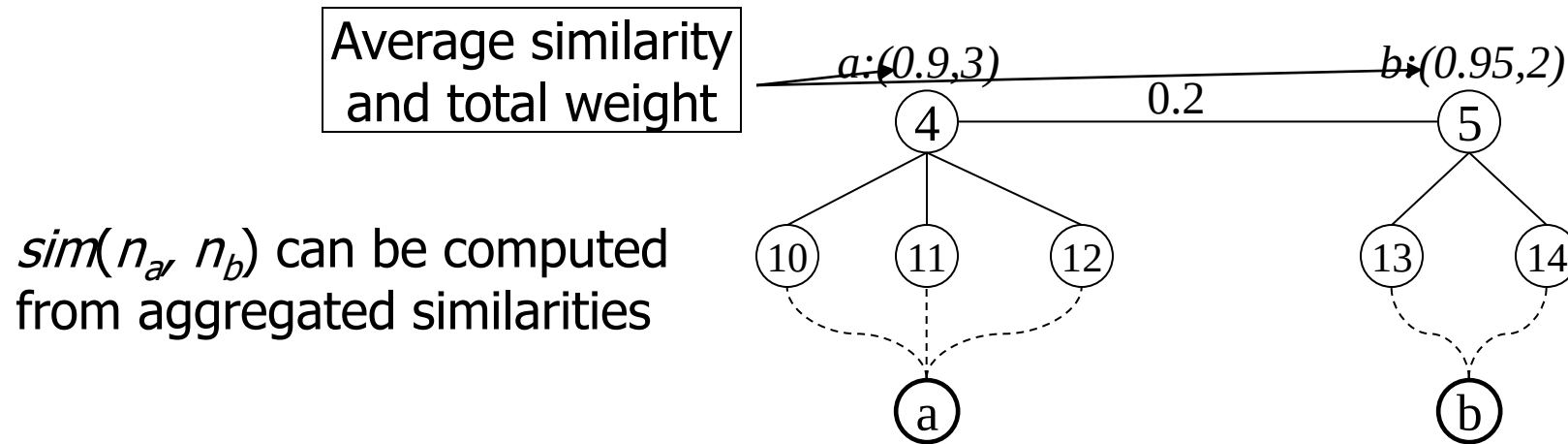
For each node $n_k \in \{n_{10}, n_{11}, n_{12}\}$ and $n_l \in \{n_{13}, n_{14}\}$, their path-based similarity $sim_p(n_k, n_l) = s(n_k, n_4) \cdot s(n_4, n_5) \cdot s(n_5, n_l)$.

$$sim(n_a, n_b) = \frac{\sum_{k=10}^{12} s(n_k, n_4)}{3} \cdot s(n_4, n_5) \cdot \frac{\sum_{l=13}^{14} s(n_l, n_5)}{2} = 0.171$$

takes $O(3+2)$ time

After aggregation, we reduce quadratic time computation to linear time computation.

Computing Similarity with Aggregation



$$sim(n_a, n_b) = avg_sim(n_a, n_4) \times s(n_4, n_5) \times avg_sim(n_b, n_5)$$

$$= 0.9 \times 0.2 \times 0.95 = 0.171$$

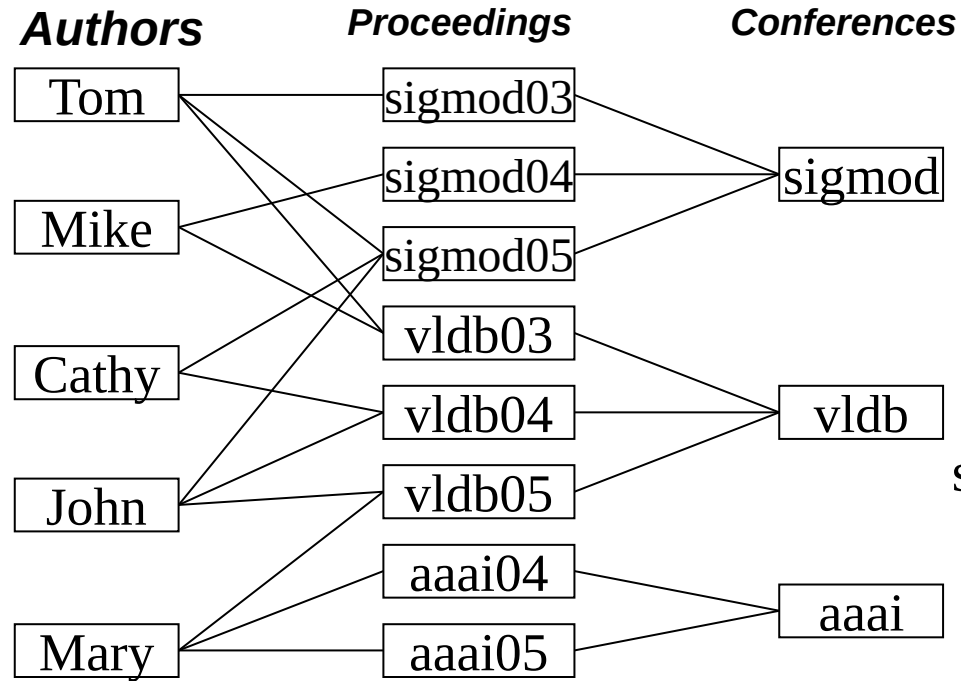
To compute $sim(n_a, n_b)$:

- Find all pairs of sibling nodes n_i and n_j , so that n_a linked with n_i and n_b with n_j .
- Calculate similarity (and weight) between n_a and n_b w.r.t. n_i and n_j .
- Calculate weighted average similarity between n_a and n_b w.r.t. all such pairs.

Chapter 10. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Overview of Clustering Methods
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods 
- Grid-Based Methods
- Summary

Link-Based Clustering: Calculate Similarities Based On Links



Jeh & Widom, KDD'2002: *SimRank*
Two objects are similar if they are linked with the same or similar objects

- The similarity between two objects x and y is defined as the average similarity between objects linked with x and those with y :

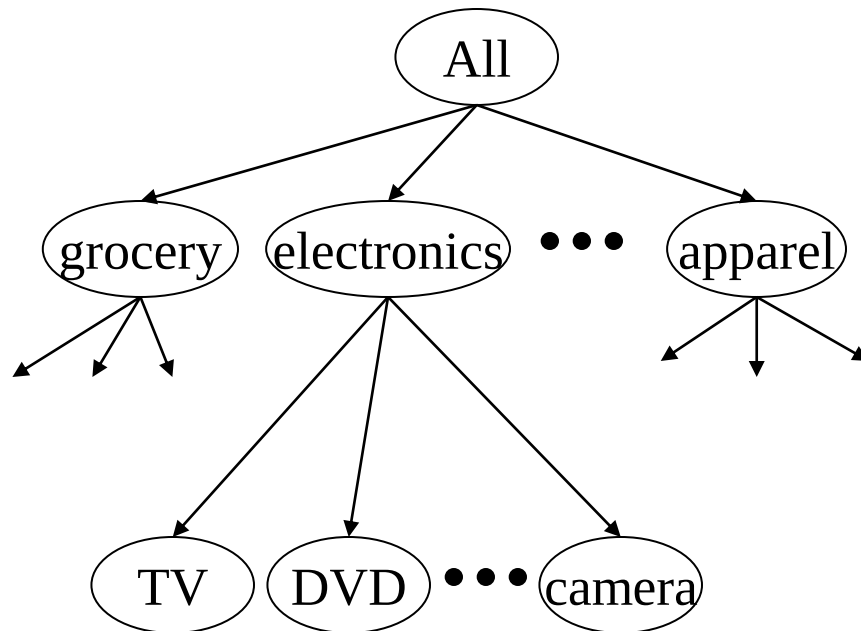
$$\text{sim}(a, b) = \frac{C}{|I(a)||I(b)|} \sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} \text{sim}(I_i(a), I_j(b))$$

- Issue: Expensive to compute:
 - For a dataset of N objects and M links, it takes $O(N^2)$ space and $O(M^2)$ time to compute all similarities.

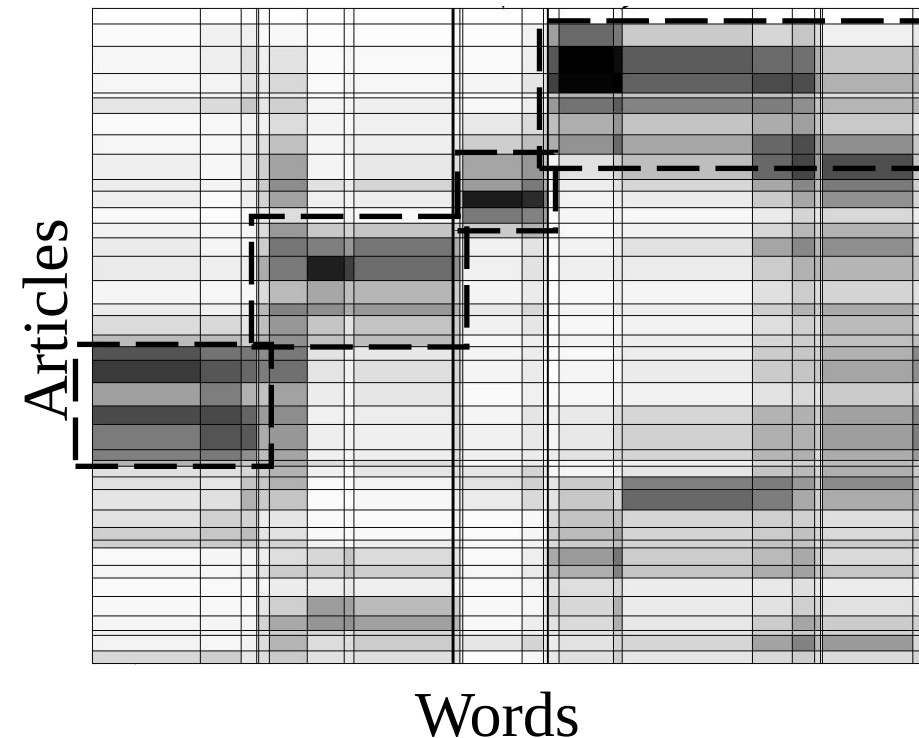
Observation 1: Hierarchical Structures

- Hierarchical structures often exist naturally among objects (e.g., taxonomy of animals)

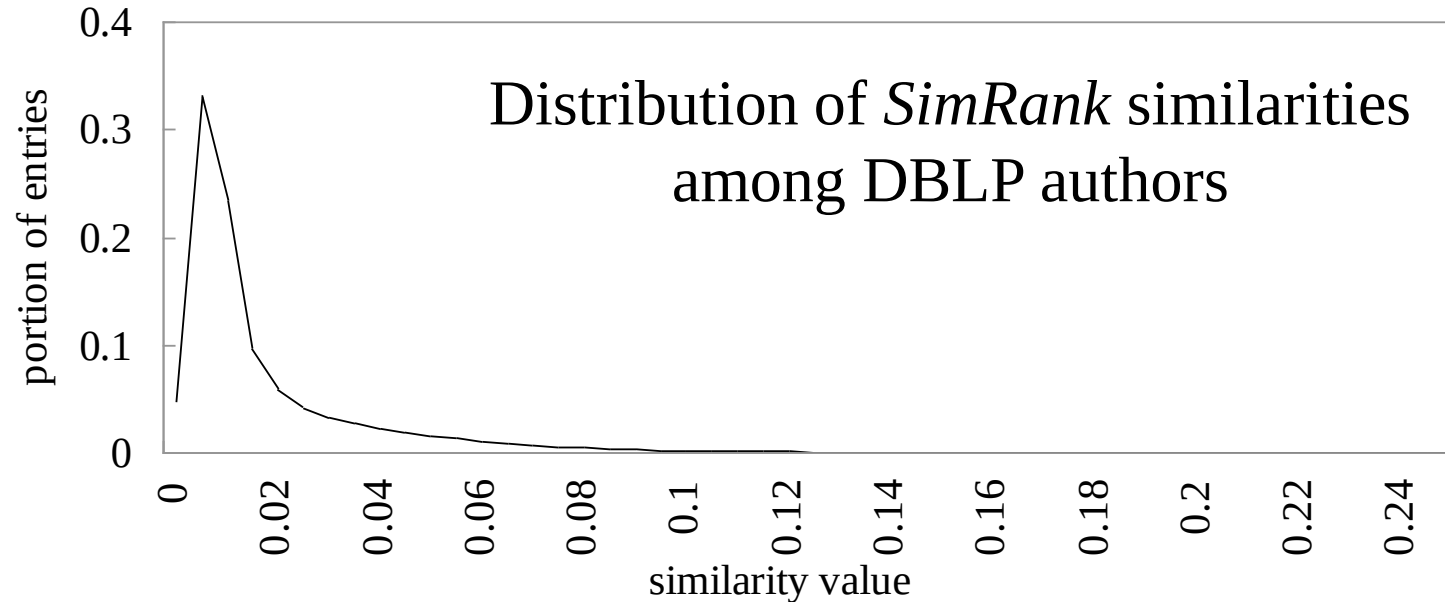
A hierarchical structure of products in Walmart



Relationships between articles and words (Chakrabarti, Papadimitriou, Modha, Faloutsos, 2004)

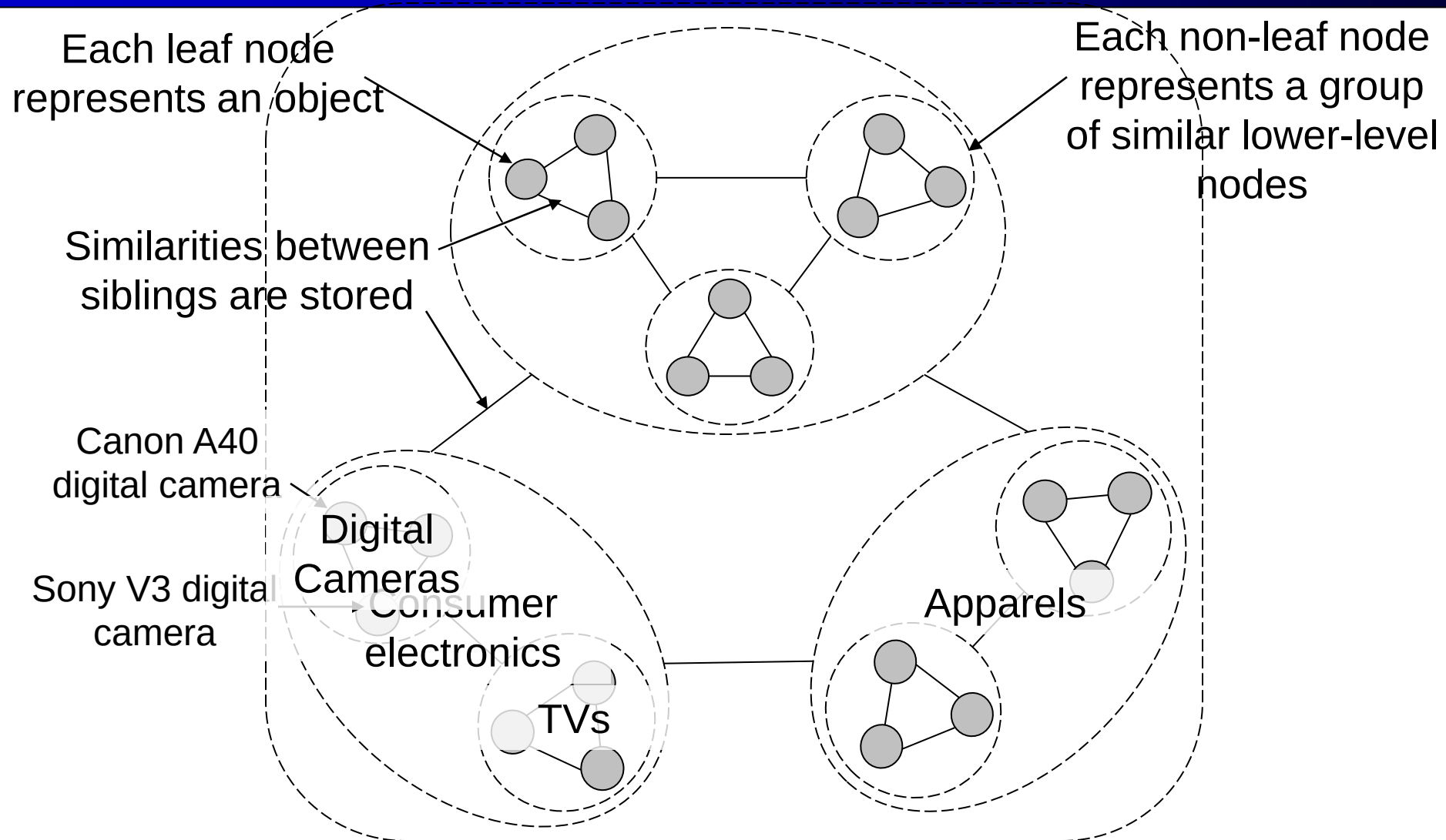


Observation 2: Distribution of Similarity

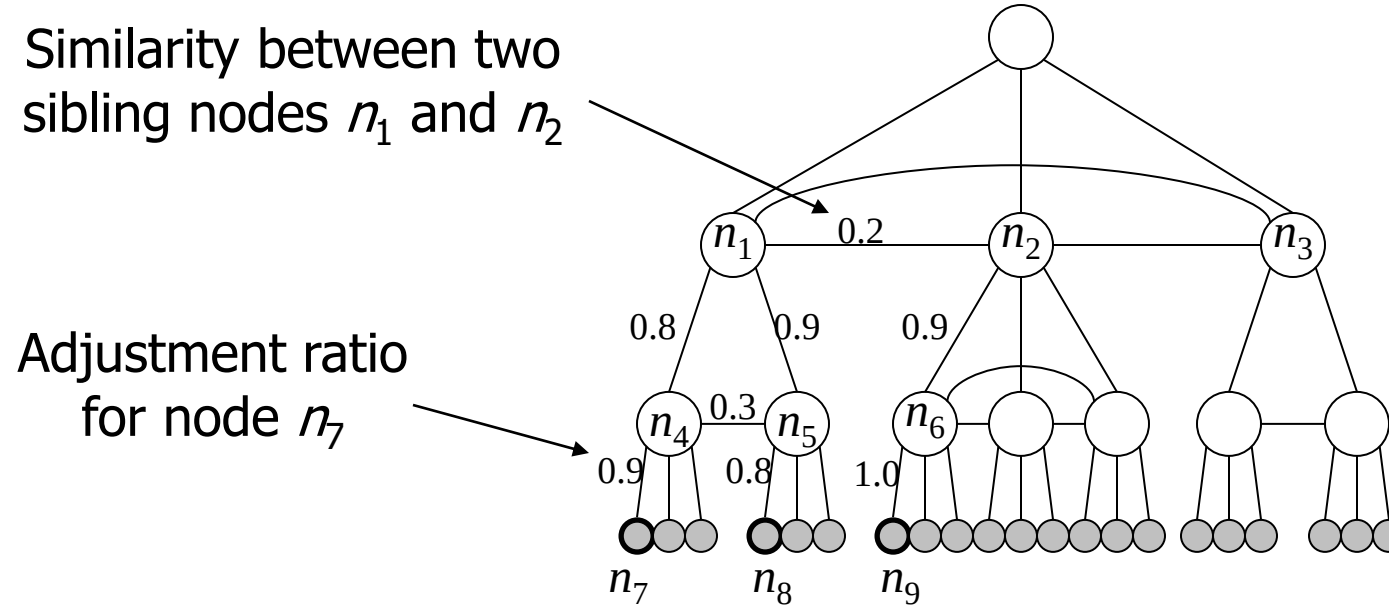


- Power law distribution exists in similarities
 - 56% of similarity entries are in $[0.005, 0.015]$
 - 1.4% of similarity entries are larger than 0.1
 - Can we design a data structure that stores the significant similarities and compresses insignificant ones?

A Novel Data Structure: SimTree



Similarity Defined by SimTree



- Path-based node similarity
 - $sim_p(n_7, n_8) = s(n_7, n_4) \times s(n_4, n_5) \times s(n_5, n_8)$
- Similarity between two nodes is the average similarity between objects linked with them in other SimTrees
- Adjust/ ratio for $x = \frac{\text{Average similarity between } x \text{ and all other nodes}}{\text{Average similarity between } x\text{'s parent and all other nodes}}$

LinkClus: Efficient Clustering via Heterogeneous Semantic Links

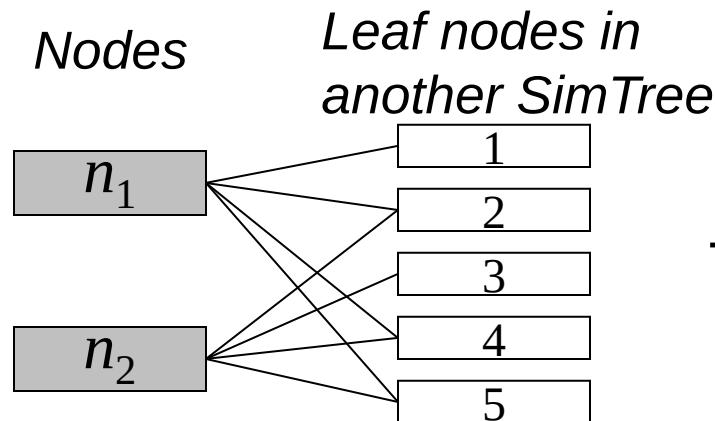
Method

- Initialize a SimTree for objects of each type
- Repeat until stable
 - For each SimTree, update the similarities between its nodes using similarities in other SimTrees
 - Similarity between two nodes x and y is the average similarity between objects linked with them
 - Adjust the structure of each SimTree
 - Assign each node to the parent node that it is most similar to

For details: X. Yin, J. Han, and P. S. Yu, “LinkClus: Efficient Clustering via Heterogeneous Semantic Links”, VLDB'06

Initialization of SimTrees

- Initializing a SimTree
 - Repeatedly find groups of tightly related nodes, which are merged into a higher-level node
- Tightness of a group of nodes
 - For a group of nodes $\{n_1, \dots, n_k\}$, its tightness is defined as the number of leaf nodes in other SimTrees that are connected to all of $\{n_1, \dots, n_k\}$

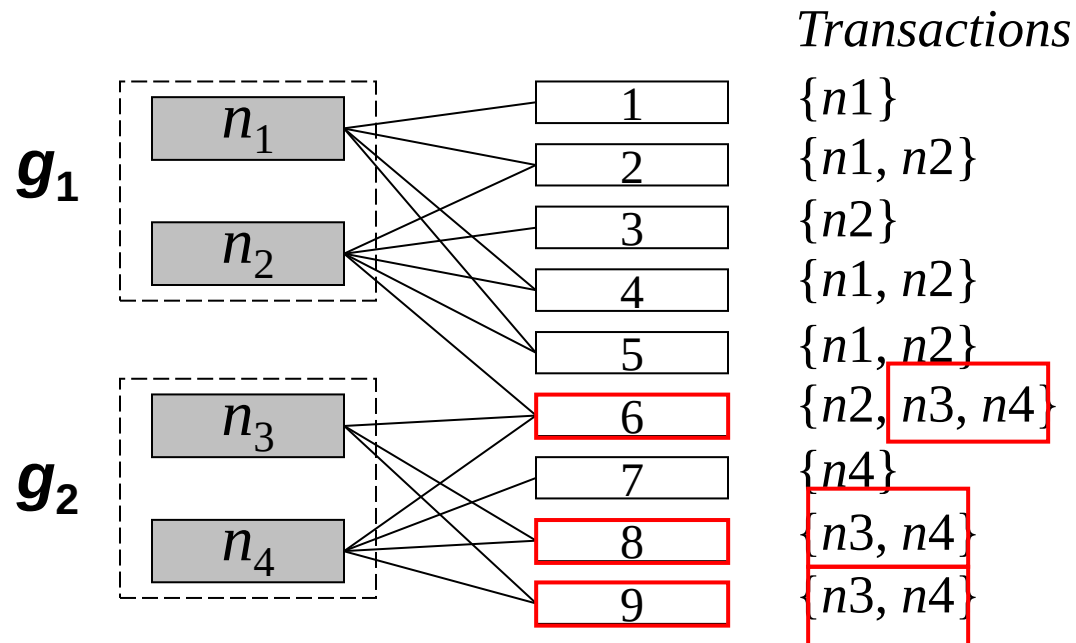


The tightness of $\{n_1, n_2\}$ is 3

Finding Tight Groups by Freq. Pattern Mining

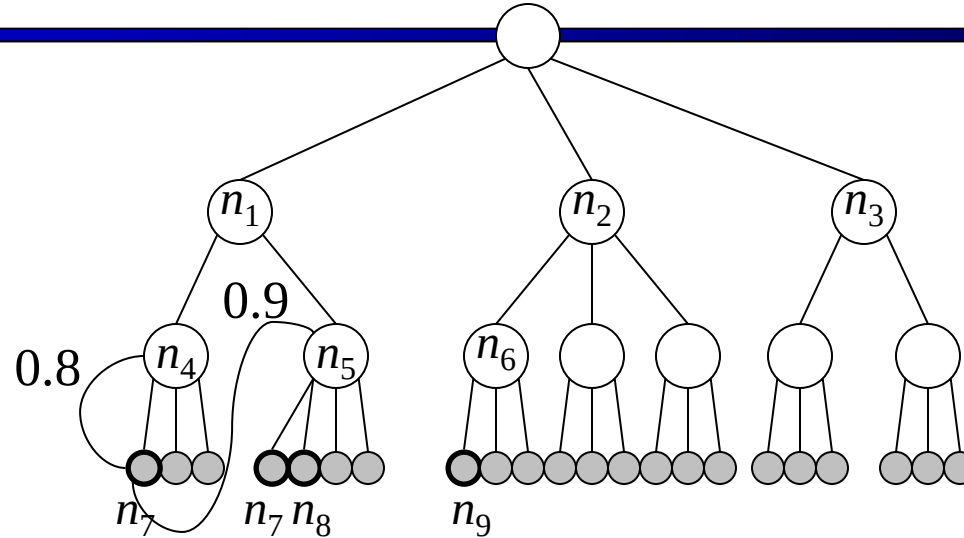
- Finding tight groups $\longrightarrow$ Frequent pattern mining
Reduced to

The tightness of a group of nodes is the support of a frequent pattern



- Procedure of initializing a tree
 - Start from leaf nodes (level-0)
 - At each level l , find non-overlapping groups of similar nodes with frequent pattern mining

Adjusting SimTree Structures



- After similarity changes, the tree structure also needs to be changed
 - If a node is more similar to its parent's sibling, then move it to be a child of that sibling
 - Try to move each node to its parent's sibling that it is most similar to, under the constraint that each parent node can have at most c children

Complexity

For two types of objects, N in each, and M linkages between them.

	Time	Space
Updating similarities	$O(M(\log N)^2)$	$O(M+N)$
Adjusting tree structures	$O(N)$	$O(N)$
<i>LinkClus</i>	$O(M(\log N)^2)$	$O(M+N)$
<i>SimRank</i>	$O(M^2)$	$O(N^2)$

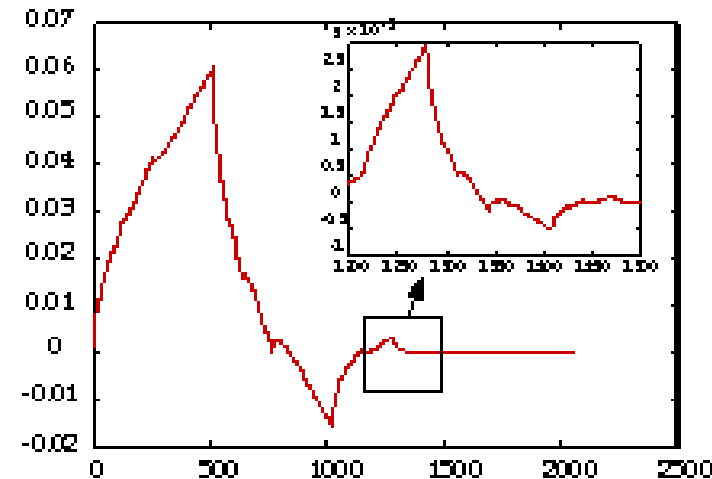
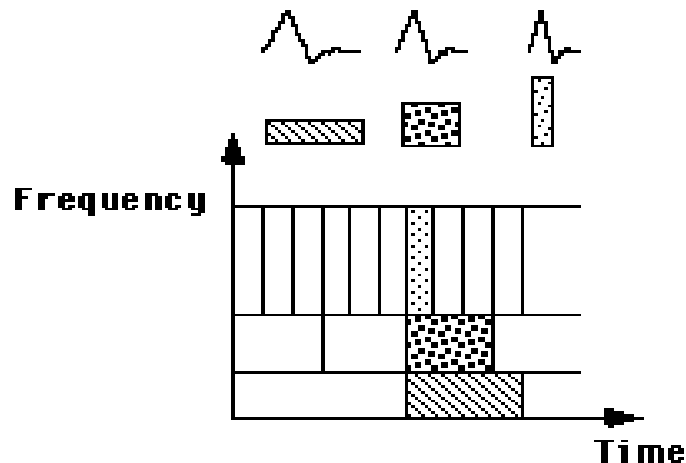
Experiment: Email Dataset

- F. Nielsen. Email dataset.
www.imm.dtu.dk/~rem/data/Email-1431.zip
- 370 emails on conferences, 272 on jobs, and 789 spam emails
- Accuracy: measured by manually labeled data
- Accuracy of clustering: % of pairs of objects in the same cluster that share common label
- Approaches compared:
 - SimRank (Jeh & Widom, KDD 2002): Computing pair-wise similarities
 - SimRank with FingerPrints (F-SimRank): Fogaras & R'acz, WWW 2005
 - pre-computes a large sample of random paths from each object and uses samples of two objects to estimate SimRank similarity
 - ReCom (Wang et al. SIGIR 2003)
 - Iteratively clustering objects using cluster labels of linked objects

<i>Approach</i>	<i>Accuracy</i>	<i>time (s)</i>
LinkClus	0.8026	1579.6
SimRank	0.7965	39160
ReCom	0.5711	74.6
F-SimRank	0.3688	479.7
CLARANS	0.4768	8.55

WaveCluster: Clustering by Wavelet Analysis (1998)

- Sheikholeslami, Chatterjee, and Zhang (VLDB'98)
- A multi-resolution clustering approach which applies wavelet transform to the feature space; both grid-based and density-based
- Wavelet transform: A signal processing technique that decomposes a signal into different frequency sub-band
 - Data are transformed to preserve relative distance between objects at different levels of resolution
 - Allows natural clusters to become more distinguishable



The WaveCluster Algorithm

- How to apply wavelet transform to find clusters
 - Summarizes the data by imposing a multidimensional grid structure onto data space
 - These multidimensional spatial data objects are represented in a n-dimensional feature space
 - Apply wavelet transform on feature space to find the dense regions in the feature space
 - Apply wavelet transform multiple times which result in clusters at different scales from fine to coarse
- Major features:
 - Complexity $O(N)$
 - Detect arbitrary shaped clusters at different scales
 - Not sensitive to noise, not sensitive to input order
 - Only applicable to low dimensional data

Quantization & Transformation

- Quantize data into m-D grid structure then wavelet transform
 - scale 1: high resolution
 - scale 2: medium resolution
 - scale 3: low resolution

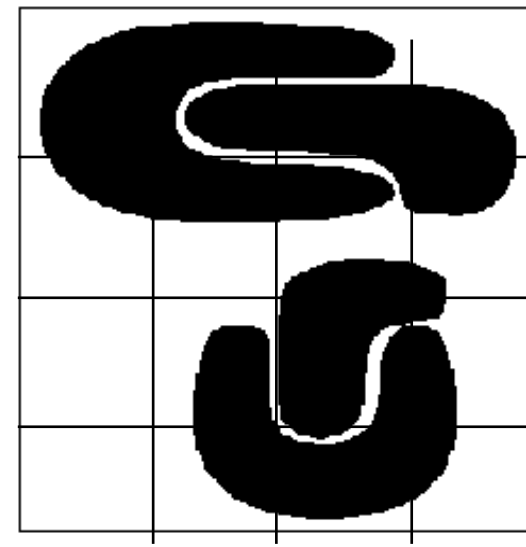
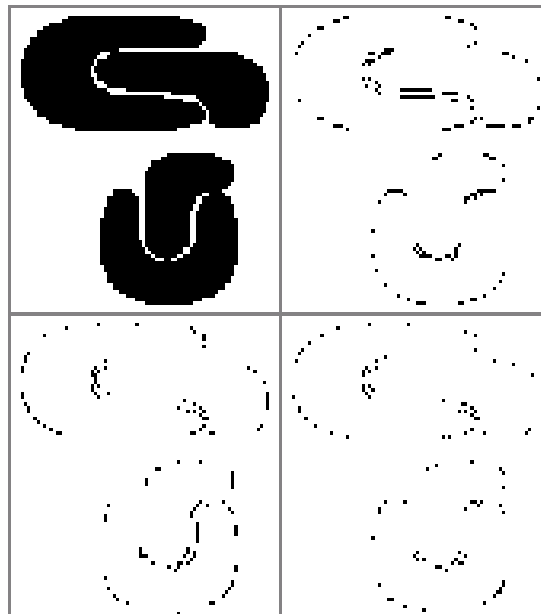


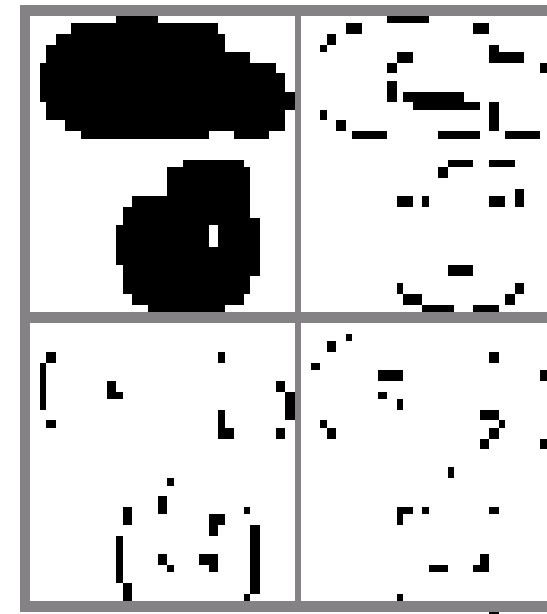
Figure 1: A sample 2-dimensional feature space.



a)



b)



c)